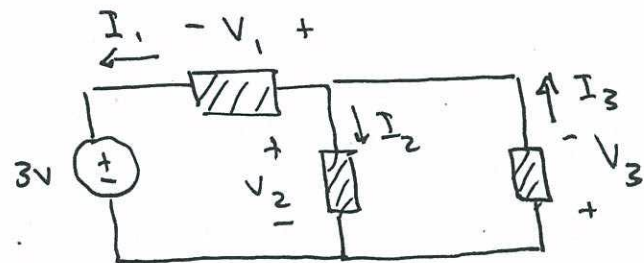
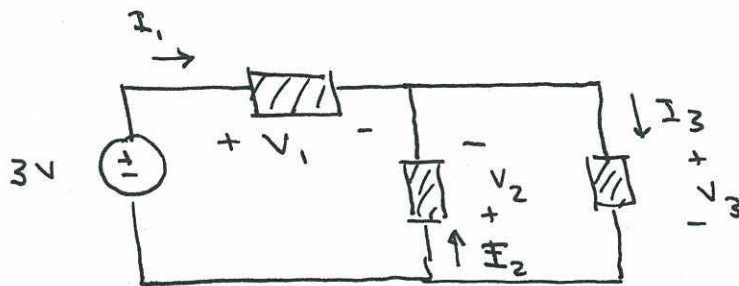
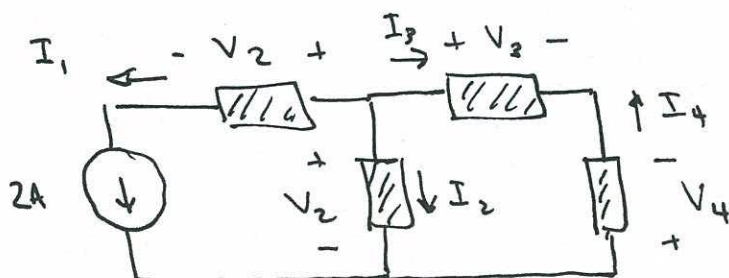
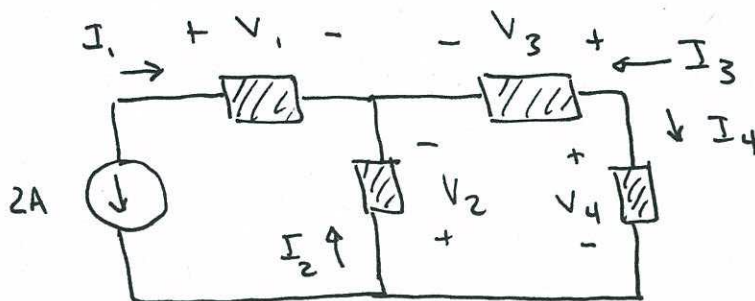


Exercises
Chapter 1.1

1(a) The only requirement is that the reference current enters the positive reference voltage terminal. Therefore, a number of solutions are correct. Two possible solutions are shown below:

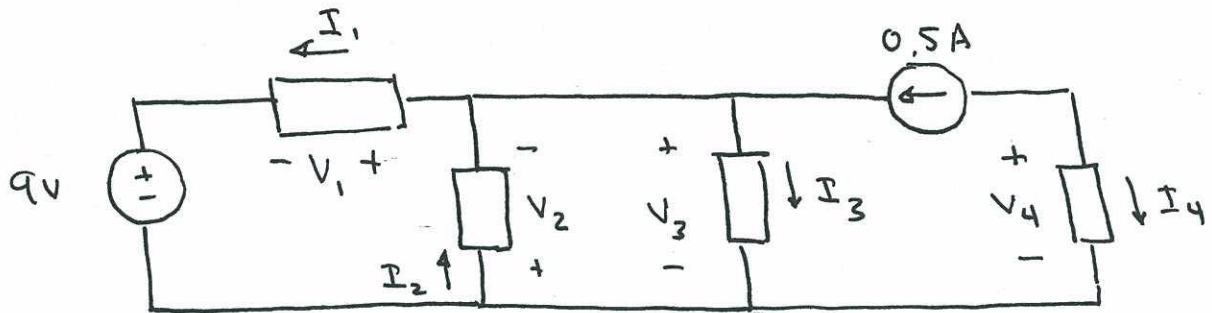


(b) Similar comments as 1(a). Two possible solutions are shown below, others are available.

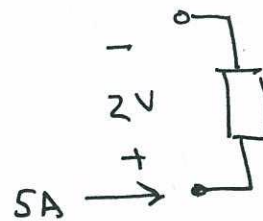
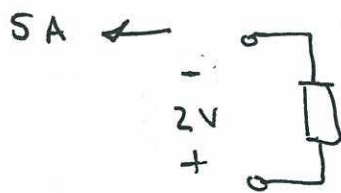


Exercises
Chapter 1.1

2. Note that current enters (assumed) positive voltage terminal, always.



3. If power is absorbed current must enter the positive voltage terminal. Both of the solutions below are correct; other solutions exist.



4. Power is supplied. current enters negative voltage terminal.

$$P = (2A)(-3V) = \underline{\underline{-6W}}$$

Exercises
Chapter 1.2

1. Can't tell what the current is, unless more is known about the circuit element. The ideal source can provide any current.
2. Can't tell, unless more is known about the circuit element. Nothing is known (directly) about the voltage across a current source.

Exercises
chapter 1.3

1. $I = \frac{V}{R} = \frac{18V}{6\Omega} \Rightarrow \underline{\underline{I = 3A}}$

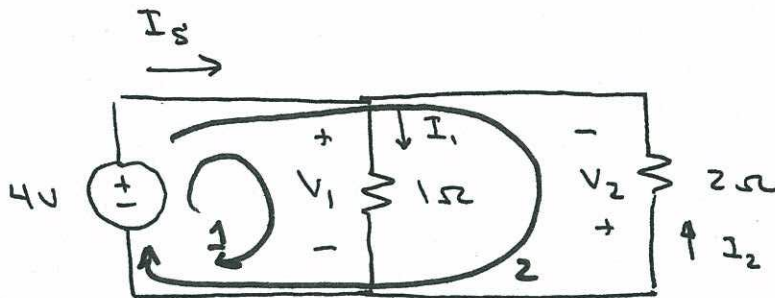
2. $V = IR = (4mA)(2k\Omega) \Rightarrow \underline{\underline{V = 8V}}$

3. $I = \frac{V}{R} = \frac{-10V}{2k\Omega} \Rightarrow \underline{\underline{I = -5mA}}$ ←

For resistors, current always enters the positive voltage node. This current must be negative.

Exercises
Chapter 1.4

1. label voltage, current reference directions



(a) KVL, Loop 2:

$$-4V - V_2 = 0 \Rightarrow V_2 = -4V \Rightarrow I_2 = \frac{V_2}{2\Omega} = \frac{-4V}{2\Omega} = -2A$$

$$\underline{I_2 = -2A} \text{ (in direction shown)}$$

(b) KVL, Loop 1:

$$-4V + V_1 = 0 \Rightarrow V_1 = 4V \Rightarrow I_1 = \frac{V_1}{R} = \frac{4V}{1\Omega} = 4A$$

$$\underline{I_1 = 4A} \text{ (positive in direction shown)}$$

(c) Total source current, I_s :

$$I_s = I_1 + (-I_2)$$

KCL; be sure to account for signs!

$$I_s = 4A + (+2A) = 6A$$

$$P_s = (4V)(-6A) = -\underline{24W} \text{ (generated)}$$

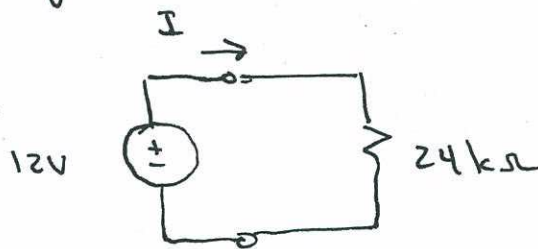
↑ our assumed signs don't agree with passive

1. Resistors are in series, so the equivalent resistance is:

$$R_{eq} = 10k\Omega + 4k\Omega + 4k\Omega + 6k\Omega$$

$$= 24k\Omega$$

Equivalent circuit:



$$I = \frac{12V}{24k\Omega} = 0.5A$$

All elements have same current, so:

$$V_1 = (4k\Omega)(0.5A) = \underline{\underline{2V}}$$

ALTERNATE: Voltage divider

$$V_1 = 12V \left(\frac{4k\Omega}{10k\Omega + 4k\Omega + 4k\Omega + 6k\Omega} \right) = 12V \left(\frac{4k\Omega}{24k\Omega} \right)$$

$$\underline{\underline{V_1 = 2V}}$$

Exercises

chapter 2.2

1. Easiest approach is probably a current divider:

$$I = 5 \text{ mA} \left(\frac{3 \text{ k}\Omega}{3 \text{ k}\Omega + 5 \text{ k}\Omega} \right) = 5 \text{ mA} \left(\frac{3}{8} \right)$$

$$\underline{\underline{I = \frac{15}{8} \text{ mA}}}$$

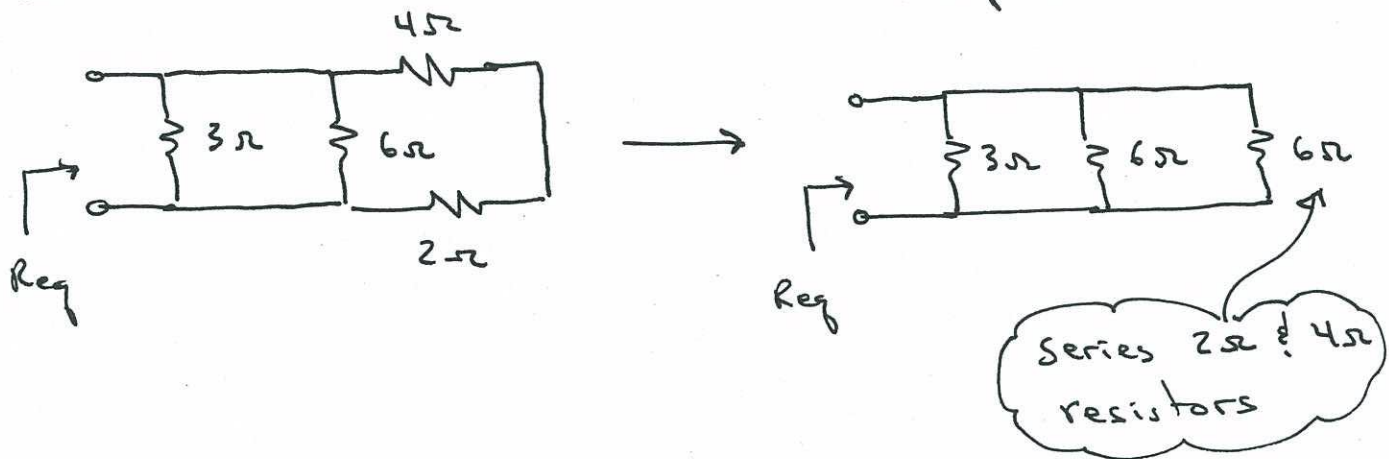
2. Probably easiest to use current divider formula:

$$I = 3 \text{ mA} \left(\frac{R}{R + 1 \text{ k}\Omega} \right) = 2 \text{ mA}$$

$$\frac{R}{R + 1 \text{ k}\Omega} = \frac{2}{3} \quad \Rightarrow \quad \underline{\underline{R = 2 \text{ k}\Omega}}$$

Exercises
Chapter 2.3

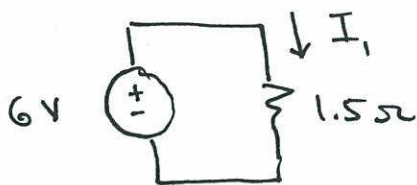
1(a)



Combine parallel 3Ω, 6Ω, 6Ω resistors:

$$R_{eq} = \frac{1}{\frac{1}{3\Omega} + \frac{1}{6\Omega} + \frac{1}{6\Omega}} = \underline{\underline{1.5\Omega}}$$

(b) From the above, an equivalent circuit is:

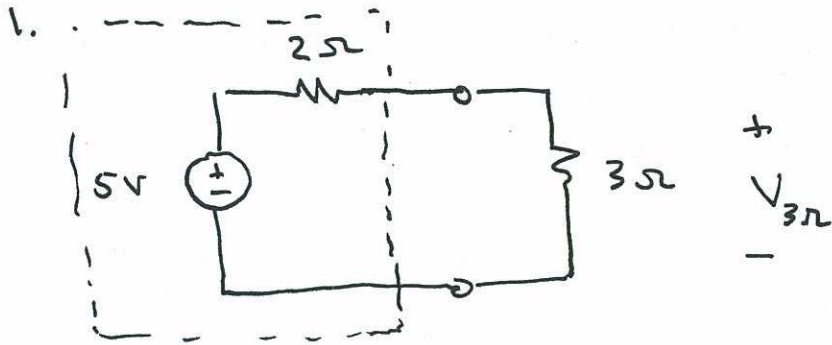


$$I_1 = \frac{6V}{1.5\Omega} = \underline{\underline{4A}}$$

could use current divider & I_1 , above to get I_2 , but probably easier to use Ohm's law:

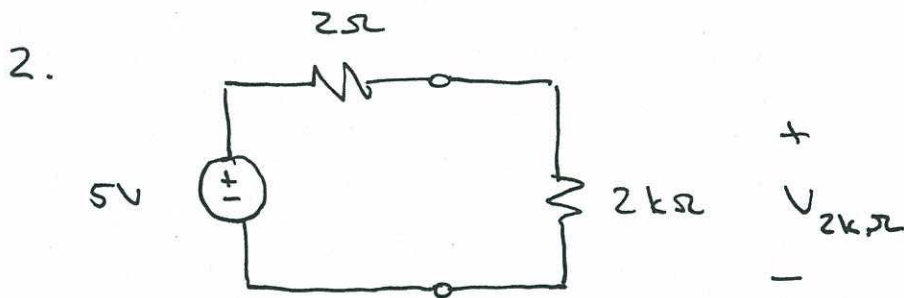
$$I_2 = \frac{6V}{3\Omega} \Rightarrow \underline{\underline{I_2 = 2A}}$$

Exercises
Chapter 2.4



voltage divider:

$$V_{3\Omega} = 5V \left(\frac{3\Omega}{2\Omega + 3\Omega} \right) = \underline{\underline{3V}}$$

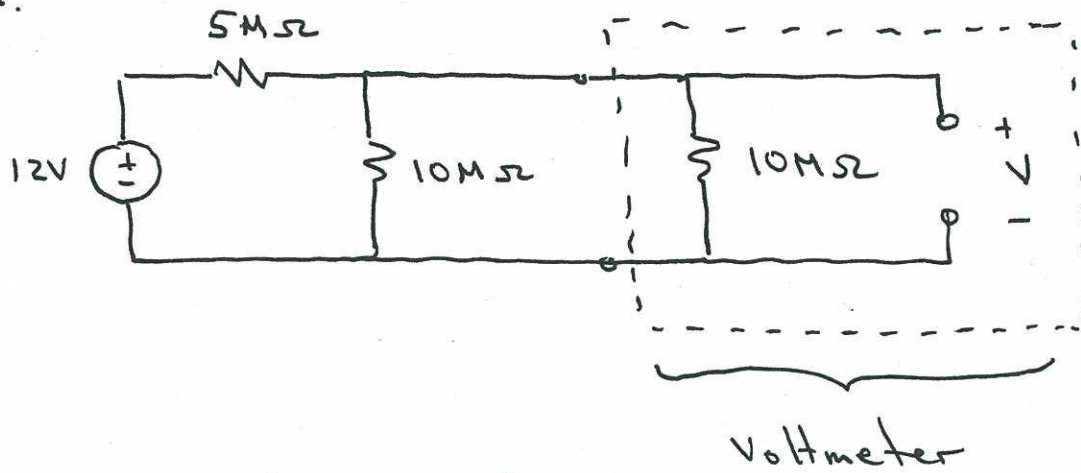


voltage divider:

$$V_{2k\Omega} = 5V \left(\frac{2000\Omega}{2000\Omega + 2\Omega} \right) = \underline{\underline{4.995V}}$$

Exercises
chapter 2.5

1.



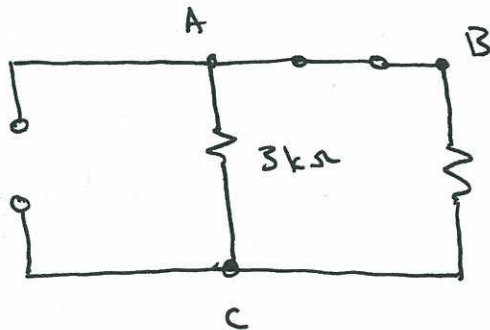
Equivalent circuit



$$V = \left(\frac{5M\Omega}{5M\Omega + 5M\Omega} \right) 12V \Rightarrow \underline{\underline{V = 6V}}$$

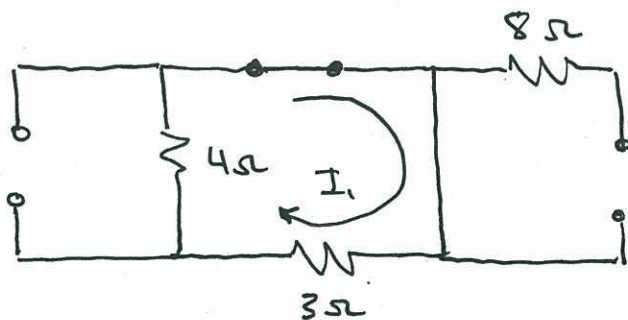
Exercises Chapter 3.1

1. Kill sources to find independent nodes:

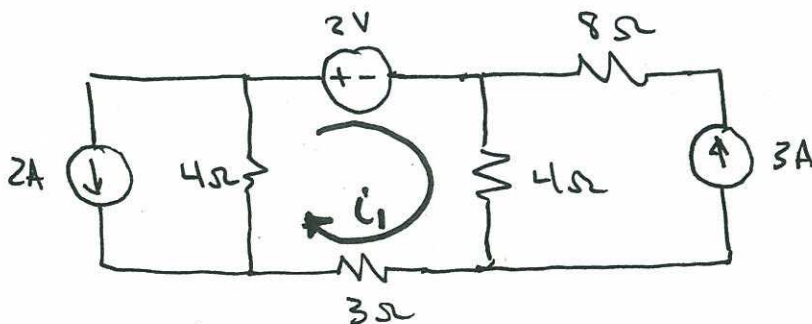


Nodes A & B are dependent. Their voltage difference is set by the 3V source.

2. Kill sources to find meshes:

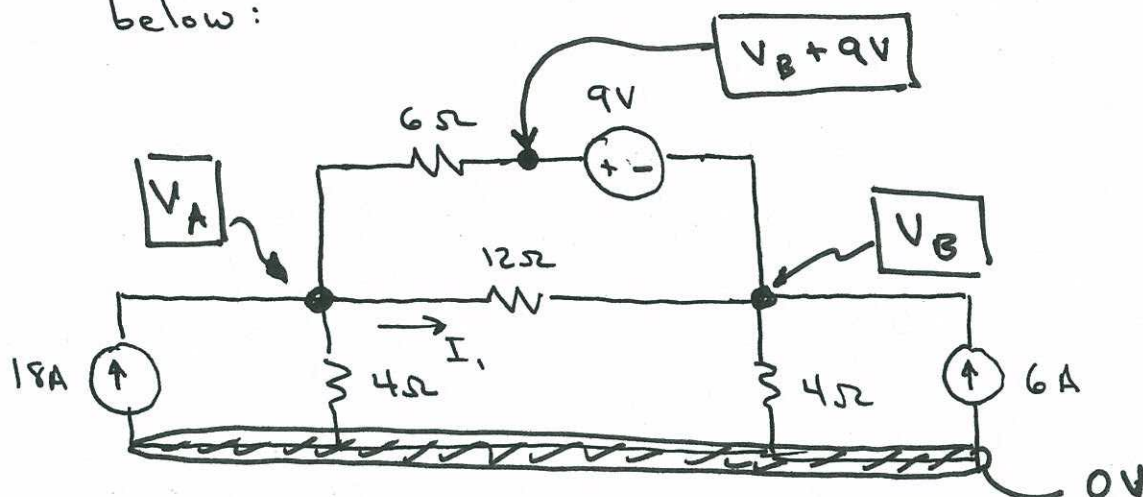


only one enclosed area, so only one mesh.



Exercises
Chapter 3.2

1. Node voltages & dependent voltages shown below:



$$\text{KCL at A: } -18\text{A} + \frac{V_A - 0\text{V}}{4\Omega} + \frac{V_A - V_B}{12\Omega} + \frac{V_A - (V_B + 9\text{V})}{6\Omega} = 0$$

$$\text{KCL at B: } -6\text{A} + \frac{V_B - 0\text{V}}{4\Omega} + \frac{V_B - V_A}{12\Omega} + \frac{(V_B + 9\text{V}) - V_A}{6\Omega} = 0$$

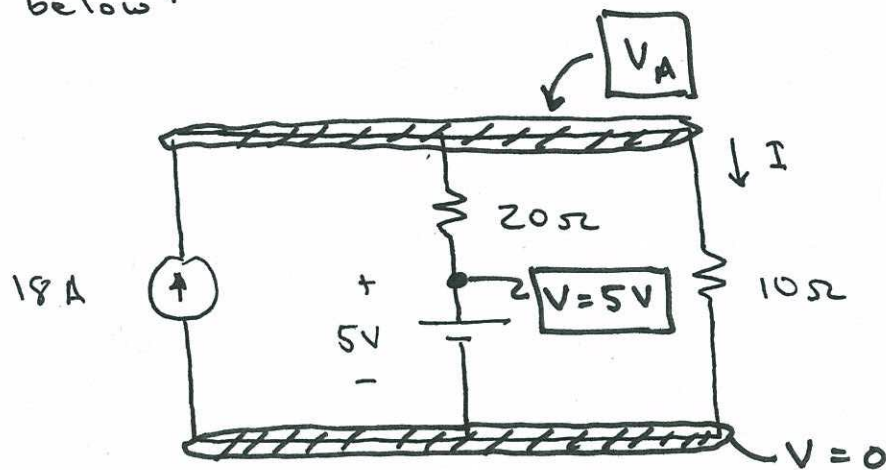
The above are two equations in two unknowns.

I_1 can be found from V_A & V_B :

$$I_1 = \frac{V_A - V_B}{12\Omega}$$

Exercises
Chapter 3.2

2. Node voltages, dependent voltages shown below:



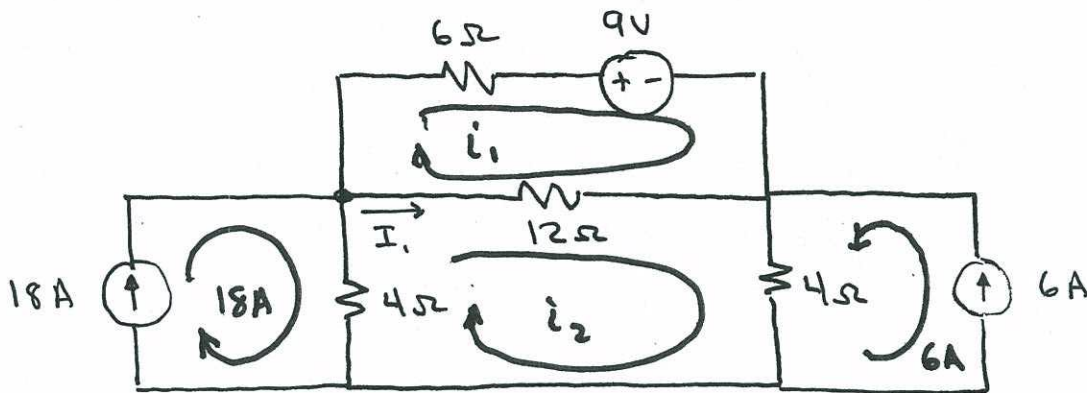
KCL at A:

$$-18\text{ A} + \frac{V_A - 5\text{ V}}{20\Omega} + \frac{V_A - 0\text{ V}}{10\Omega} = 0 \Rightarrow V_A = \frac{365}{3}\text{ V}$$

$$I = \frac{V_A}{10\Omega} \Rightarrow \underline{\underline{I = \frac{365}{30}\text{ A}}}$$

Exercises
chapter 3.3

1. Mesh loops, constrained loops shown below:



KVL, Loop i_1 :

$$6\Omega (i_1) + 9V + 12\Omega (i_1 - i_2) = 0 \quad (1)$$

KVL, Loop i_2 :

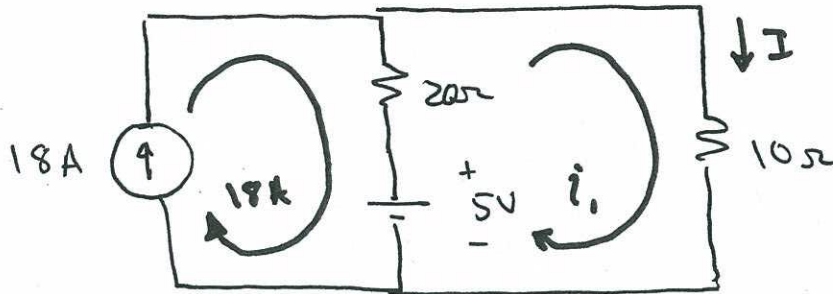
$$4\Omega (i_2 - 18A) + 12\Omega (i_2 - i_1) + 4\Omega (i_2 + 6A) = 0 \quad (2)$$

$$I_1 = i_2 - i_1 \quad (3)$$

3 equations, 3 unknowns!

Exercises
chapter 3.4

2. Mesh i_1 constrained
loops shown below:



KVL, Loop i_1 :

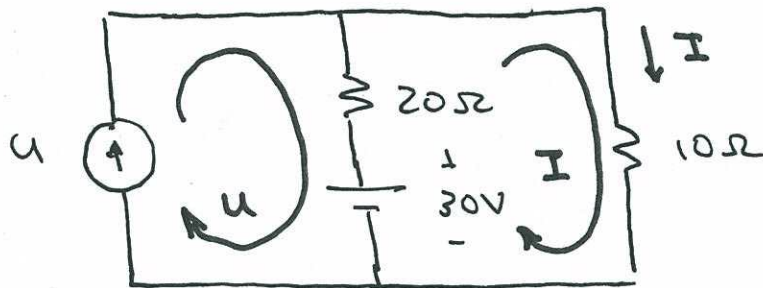
$$-5V + 20\Omega (i_1 - 18A) + 10\Omega (i_1) = 0$$

$$i_1 = \frac{365}{30} A$$

$$I = i_1 \Rightarrow \underline{\underline{I = \frac{365}{30} A}}$$

Exercises
chapter 4.1

1. Rather arbitrarily, use mesh analysis.
Mesh & dependent loops shown below:

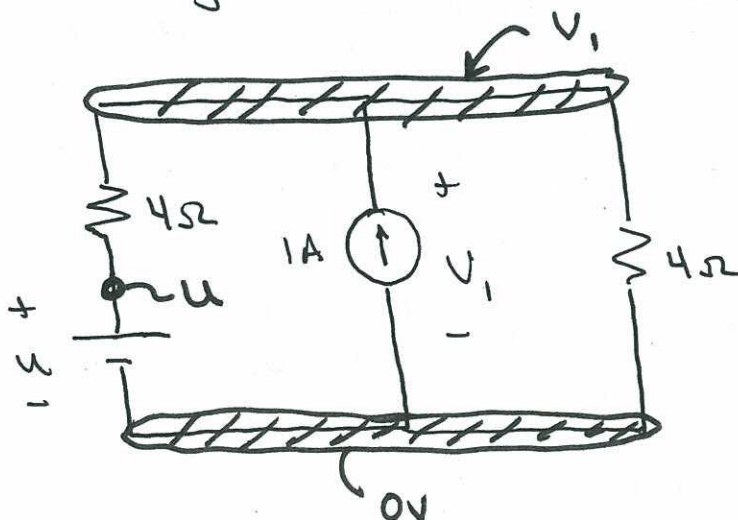


$$\text{KVL, Loop I: } -30\text{V} + 20\Omega (I - u) + 10\Omega (I) = 0$$

$$30 + 20u = 30I$$

$$\underline{\underline{I = 1 + \frac{2}{3}u}}$$

2. Use nodal analysis. Node voltages & dependent voltages shown below:



$$\underline{\text{KCL, node } V_1:}$$

$$\frac{V_1 - u}{4\Omega} - 1\text{A} + \frac{V_1}{4\Omega} = 0$$

$$2V_1 = u + 4$$

$$\underline{\underline{V_1 = \frac{u + 4}{2}}}$$

Exercises
Chapter 4.2

1.

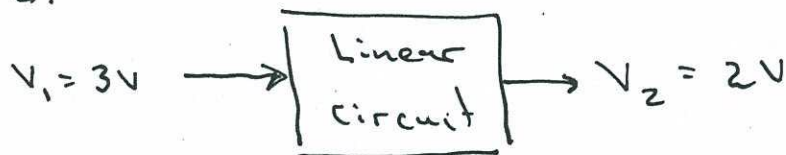
$$(a) \quad V_a = 20\Omega (2A) = 40V$$

$$(b) \quad V_b = 20\Omega (3A) = 60V$$

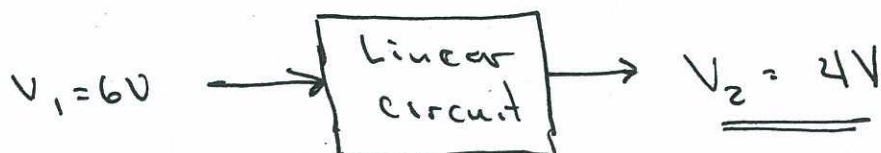
$$(c) \quad V_c = 20\Omega (5A) = 100V$$

$V_c = V_a + V_b \Rightarrow$ looks like superposition holds!

2.



$\downarrow$ multiply input by 2, so output
also scaled by 2

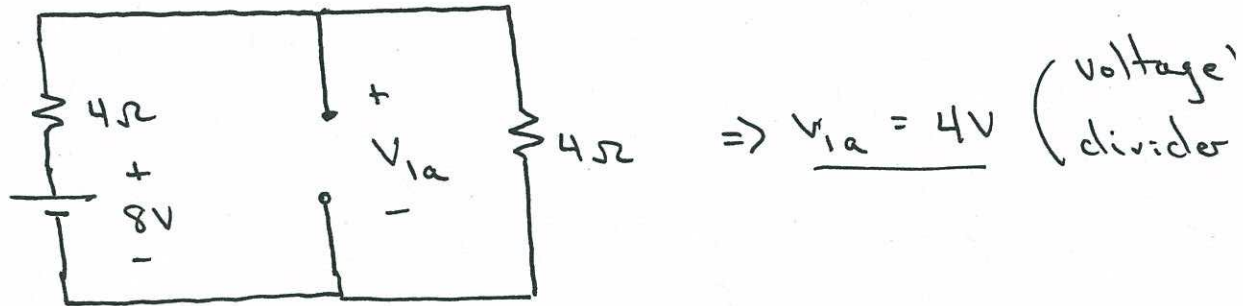


Exercises

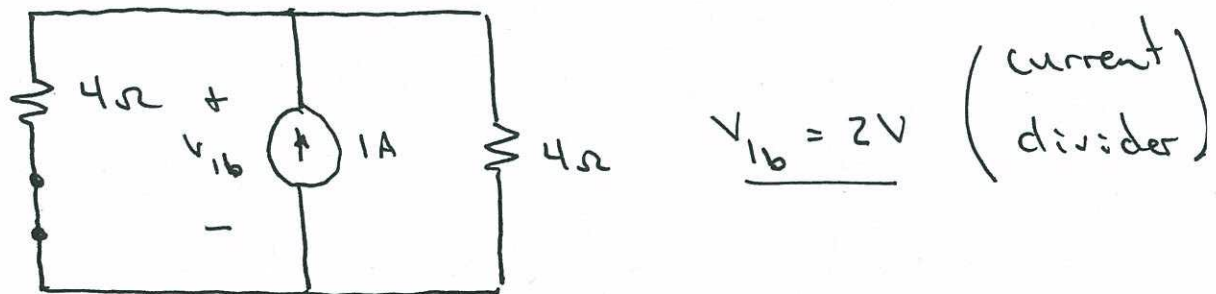
Chapter 4.3

1.

(a) Contribution from 8V source:



(b) Contribution from 1A source:



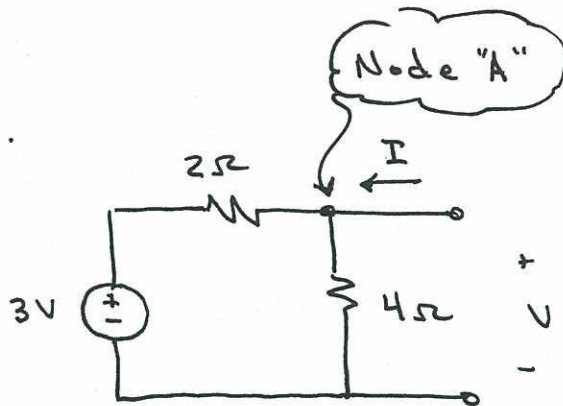
$$V_1 = V_{1a} + V_{1b} = 4V + 2V$$

$$\underline{\underline{V_1 = 6V}}$$

Exercises

Chapter 4.4

1.



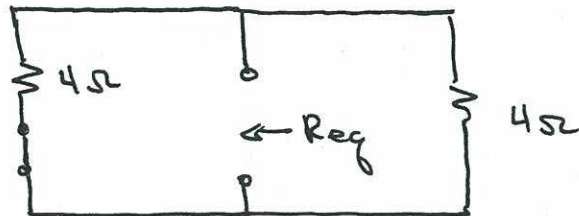
$$\text{KCL at A: } I = \frac{V}{4\Omega} + \frac{V - 3V}{2\Omega}$$

$$\underline{\underline{4I = 3V - 6}} \quad \leftarrow \text{(or any other equivalent mathematical statement)}$$

Exercises

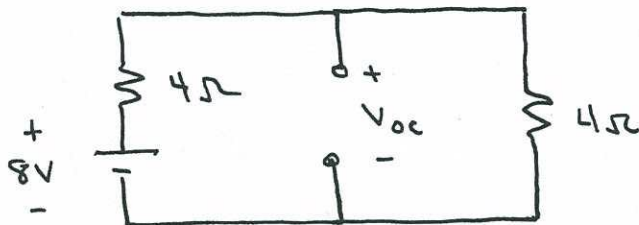
Chapter 4.5

1. Find R_{Th} :



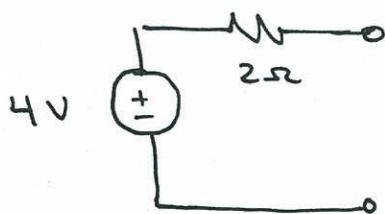
$$R_{Th} = 2\Omega$$

Find V_{oc} :

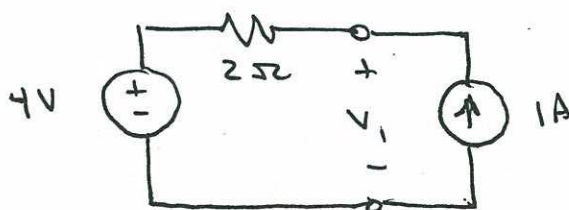


$$V_{oc} = 4V$$

Thevenin circuit:



To find V_1 :

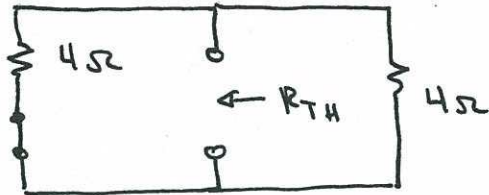


$$V_1 = 4V + 2\Omega(1A)$$

$$\underline{\underline{V_1 = 6V}}$$

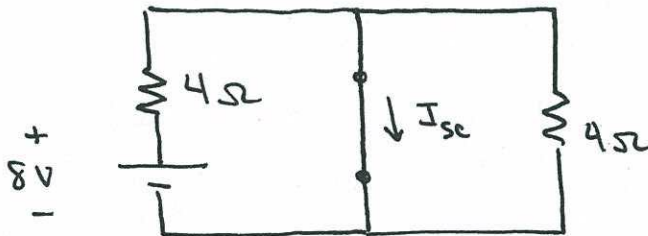
Exercises
Chapter 4.5

2. Find R_{TH} :



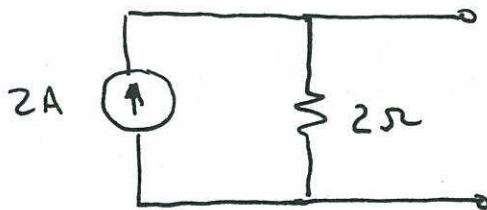
$$R_{TH} = 2\Omega$$

Find I_{sc}

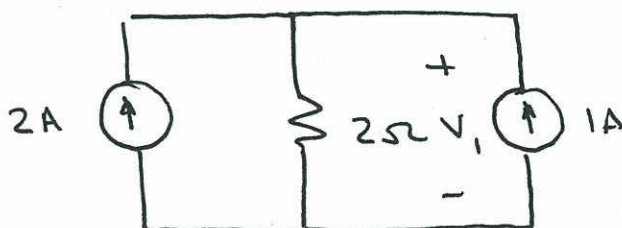


$$I_{sc} = 2A$$

Norton circuit:



Find V_1 :



$$V_1 = (3A)(2\Omega)$$

$$\underline{\underline{V_1 = 6V}}$$

Exercises

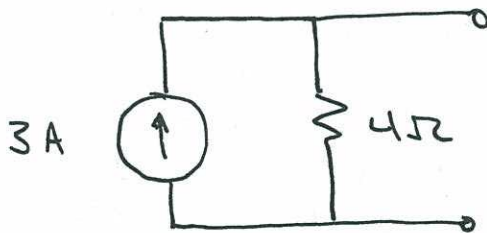
Chapter 4.5

3.

We are given (basically) a
Thevenin circuit. To convert to a

Norton circuit, $I_{sc} = \frac{V_{oc}}{R_{TH}} = \frac{12V}{4\Omega} = 3A$

The Norton circuit is:



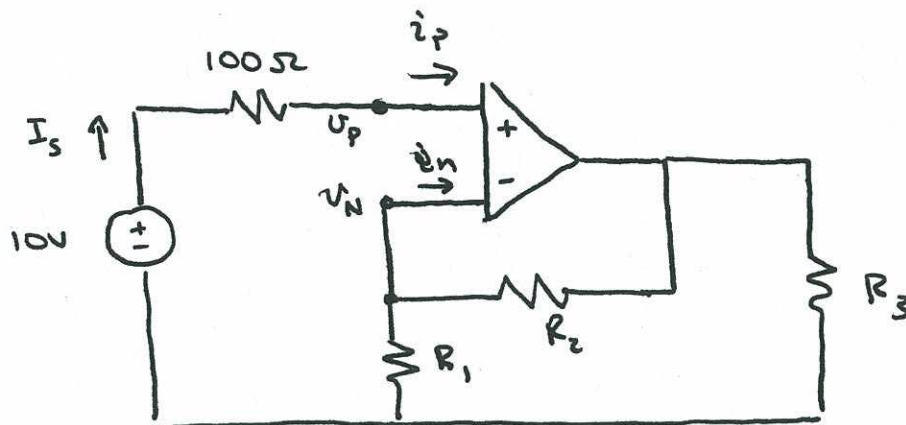
Exercises
Chapter 4.6

1. The circuit to the left of the resistor R is a Thevenin circuit. Therefore, for maximum power, set:

$$R = R_{TH} = \underline{\underline{24\Omega}}$$

Exercises
chapter 5.1

1.



Op-amp rules for the above circuit:

(1) $v_P = v_N$

(2) $i_P = i_N = 0$

(a) Since $i_P = 0 \Rightarrow \underline{\underline{I_s = 0}}$ (they are in series)

(b) Since $i_s = 0 \Rightarrow$ there is no current through the 100Ω resistor $\hat{=}$ no voltage drop across the 100Ω resistor $\Rightarrow \underline{\underline{v_P = 10V}}$

(c) Since $v_P = v_N$, and from (b):

$$\underline{\underline{v_N = 10V}}$$

Exercises

Chapter 5.2

$$K = 10,000$$

$$V^+ = 20V$$

$$V^- = -10V$$

For an op-amp, $V_{out} = K(v_p - v_n)$ but

$$V^- < V_{out} < V^+$$

$$(a) \quad v_p - v_n = 1mV \Rightarrow V_{out} = 10,000(.001) = \underline{\underline{10V}}$$

$$(b) \quad v_p - v_n = 2mV \Rightarrow V_{out} = 10,000(.002) = \underline{\underline{20V}}$$

$$(c) \quad v_p - v_n = 4mV \Rightarrow V_{out} = 10,000(.004) = 40V$$

$$\underline{\underline{BUT}} \quad V_{out} < V^+ \Rightarrow \underline{\underline{V_{out} = 20V}}$$

$$(d) \quad v_p - v_n = -0.2mV \Rightarrow V_{out} = 10,000(-0.0002) = \underline{\underline{-2V}}$$

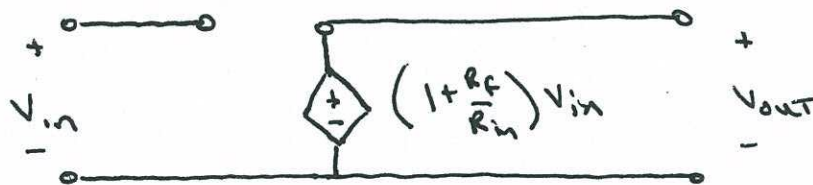
$$(e) \quad v_p - v_n = -2mV \Rightarrow V_{out} = 10,000(-0.002) = -20V$$

$$\underline{\underline{BUT}} \quad V^- < V_{out} \Rightarrow \underline{\underline{V_{out} = -10V}}$$

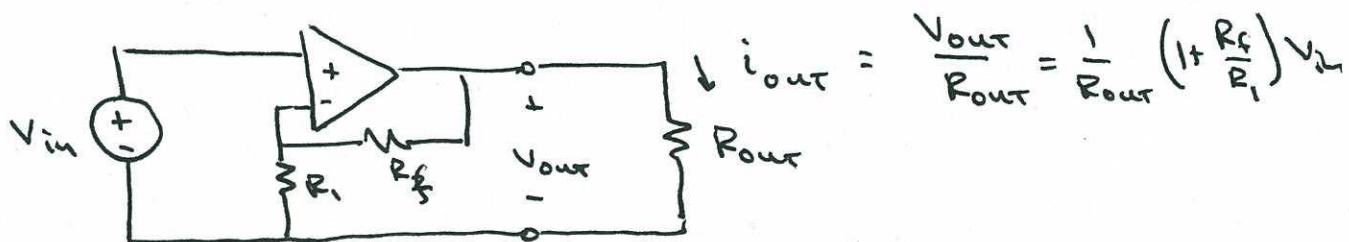
1. The circuit of Example 2 is governed by the relation:

$$V_{out} = \left(1 + \frac{R_f}{R_{in}}\right) V_{in}$$

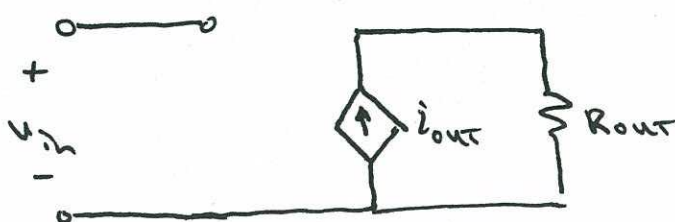
As a VCVS:



2. Since the circuit of Example 2 sets a voltage we need to have an output resistance to find the current. The circuit becomes:

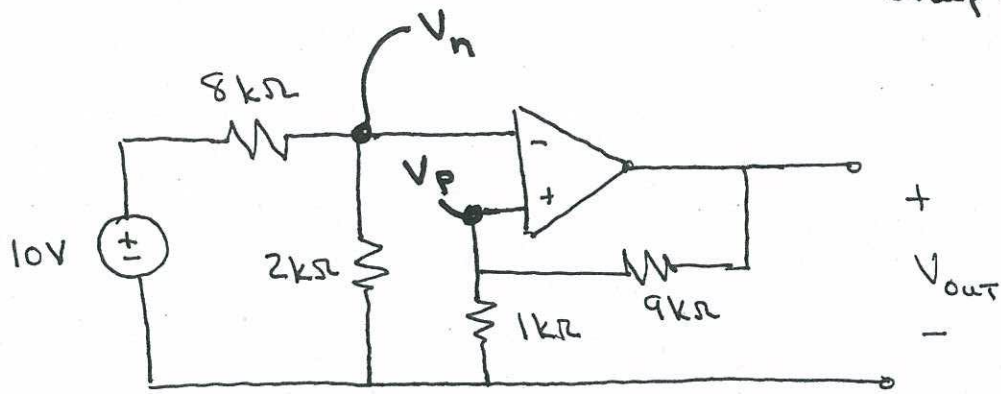


As a voltage controlled current source:



$$i_{out} = \frac{1}{R_{out}} \left(1 + \frac{R_f}{R_{in}}\right) V_{in}$$

3.



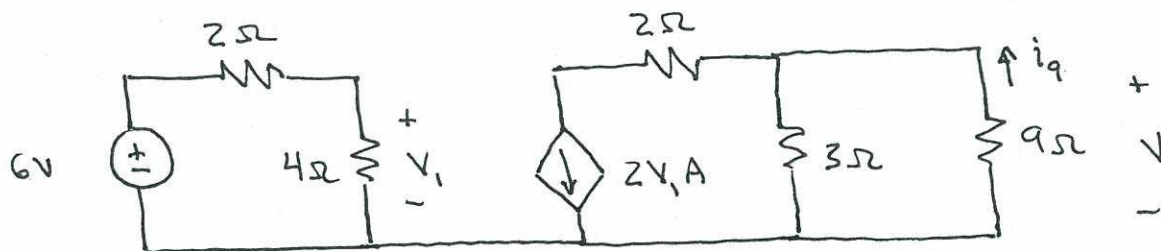
$$V_n = 10V \left[\frac{2k\Omega}{2k\Omega + 8k\Omega} \right] = 2V \quad (\text{voltage divider})$$

$$V_p = V_n = 2V \quad (\text{op-amp rules})$$

$$\text{KCL at non-inverting input: } \frac{2V - 0}{1k\Omega} = \frac{V_{out} - 2V}{9k\Omega}$$

$$\underline{\underline{V_{out} = 20V}}$$

4.



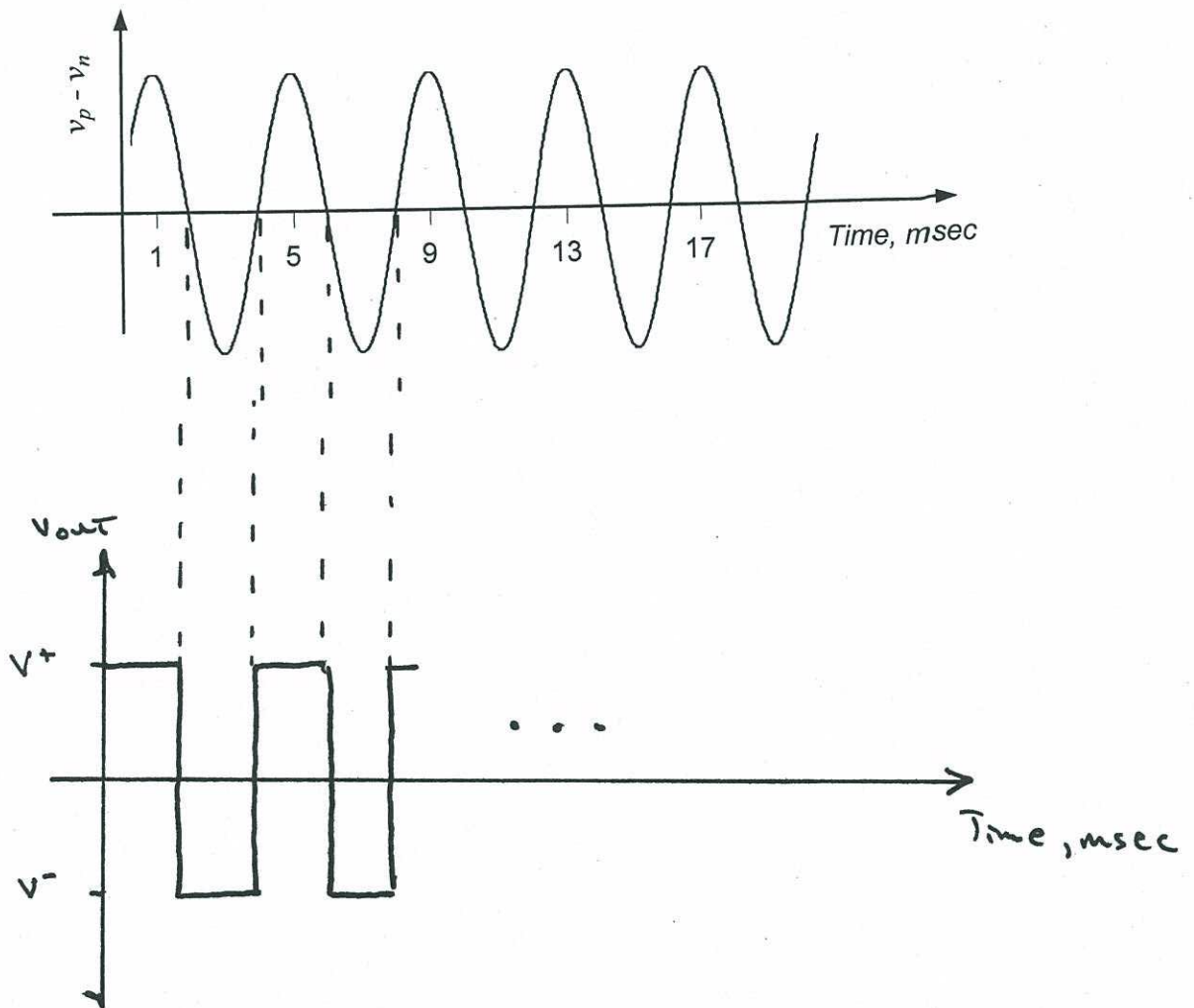
$$V_1 = 6V \left(\frac{4\Omega}{2\Omega + 4\Omega} \right) = 4V \quad (\text{voltage divider})$$

$$2V_1 = 8A \quad (\text{dependent source current})$$

$$i_q = 8A \left[\frac{3\Omega}{3\Omega + 9\Omega} \right] = 2A \quad (\text{voltage divider})$$

$$V = -i_q (9\Omega) \Rightarrow \underline{\underline{V = -18V}}$$

Exercises
Chapter 5.5



↗
The output voltage is V^+ if
 $v_p - v_n > 0$ and V^- if $v_p - v_n < 0$.

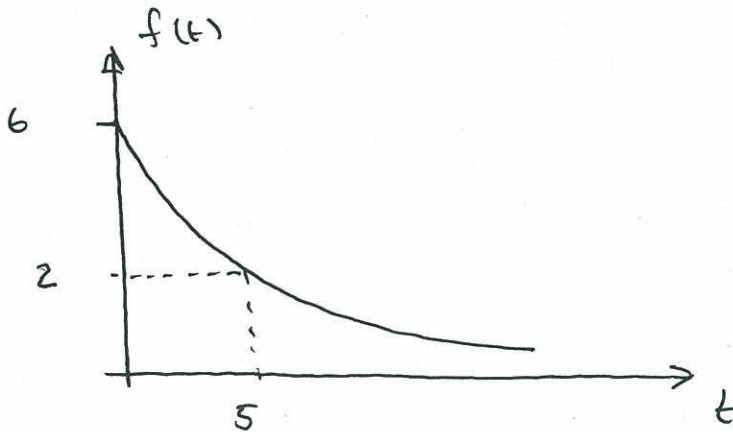
Exercises
chapter 6.1

1. The steady-state response is the response as $t \rightarrow \infty$. In this case, the velocity of the mass as $t \rightarrow \infty$ is zero (it comes to rest).

$$\underline{\underline{v(t \rightarrow \infty) = 0 \text{ m/s}}}$$

1. $f(t) = u(t+1) - 3u(t-1) + 2u(t-3)$

2.



Form of exponential: $f(t) = A_0 e^{-t/\tau}$

From graph, $A_0 \approx 6$
 $\tau \approx 5 \text{ msec}$ } $\Rightarrow f(t) \approx 6 e^{-t/5 \times 10^{-3}}$

$$\underline{\underline{f(t) \approx 6 e^{-200t}}}$$

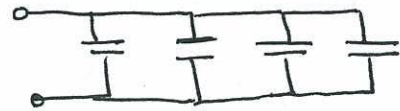
Exercises Chapter 6.3

1.

- maximum capacitance from parallel combination (capacitances add directly)

$$C_{\max} = 4 \mu F$$

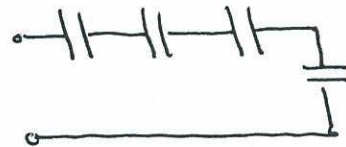
$\Rightarrow$



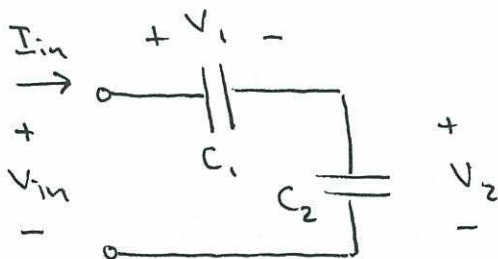
- minimum capacitance from series combination:

$$C_{\text{eq}} = 0.25 \mu F$$

$\Rightarrow$



2.



$$I_{\text{in}} = \frac{C_1 C_2}{C_1 + C_2} \frac{dV_{\text{in}}}{dt}$$

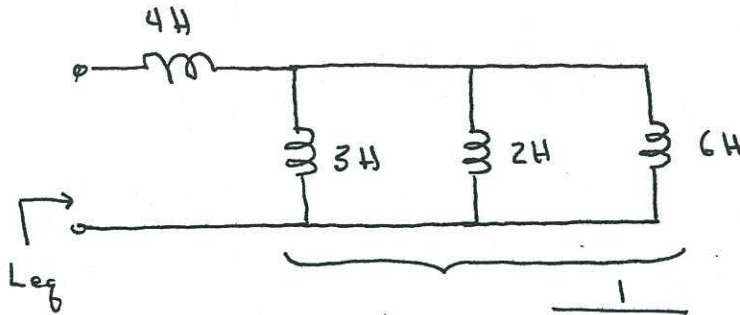
$$V_1 = \frac{1}{C_1} \int I_{\text{in}} dt + V_{\text{in}}(0) \Rightarrow \underline{\underline{V_1 = \frac{C_2}{C_1 + C_2} V_{\text{in}}}}$$

Likewise, $\underline{\underline{V_2 = \frac{C_1}{C_1 + C_2} V_{\text{in}}}}$

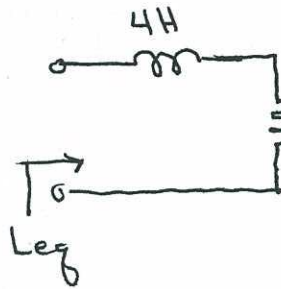
If $C_1 = C_2 = 10 \mu F$, $\underline{\underline{V_2 = \frac{1}{2} V_{\text{in}}}}$

Exercises
Chapter 6.4

1.



$$L_{eq_1} = \frac{1}{\frac{1}{3H} + \frac{1}{2H} + \frac{1}{6H}} = 4H$$



$$\Rightarrow L_{eq} = 4H + 4H = \underline{\underline{8H}}$$

Exercises
Chapter 7.1

1. (a)

i. Equation is: $mc_p \frac{d(T_B - T_0)}{dt} + h(T_B - T_0) = 0$

as $t \rightarrow \infty$, temperatures become constant,

and $\frac{d(T_B - T_0)}{dt} \rightarrow 0$. Therefore:

$$h(T_B - T_0) = 0 \Rightarrow \underline{T_B = T_0}$$

Body temperature approaches
ambient temperature as $t \rightarrow \infty$

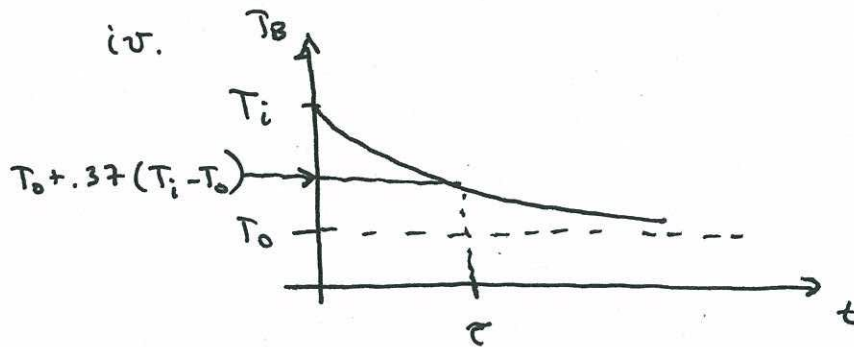
ii. To find time constant, write equation in
the form $\frac{dy}{dt} + \frac{1}{\tau} y = 0$:

$$\frac{d(T_B - T_0)}{dt} + \frac{h}{mc_p} (T_B - T_0) = 0 \Rightarrow \underline{\underline{\tau = \frac{mc_p}{h}}}$$

iii. If mass doubles, the time constant doubles.
Makes sense, since a larger mass should
take longer to cool down.

Exercises
Chapter 7.1

1(a) continued



(b)

i) Governing equation: $m c_p \frac{d(T_B - T_0)}{dt} + h(T_B - T_0) = q_{in}$

The input is constant, so all temperatures become constant as $t \rightarrow \infty$ and $\frac{d(T_B - T_0)}{dt} \rightarrow 0$:

$$h(T_B - T_0) = q_{in} \Rightarrow \underline{\underline{T_B = \frac{q_{in}}{h} + T_0 \text{ as } t \rightarrow \infty}}$$

ii. Doubling the heat input doubles the increase in body temperature (above T_0).

Reasonable - more heat gives higher temp.

iii. Doubling mass does not affect the final temperature. Mass only affects how quickly the mass can heat up & cool down.

Exercises
chapter 7.1

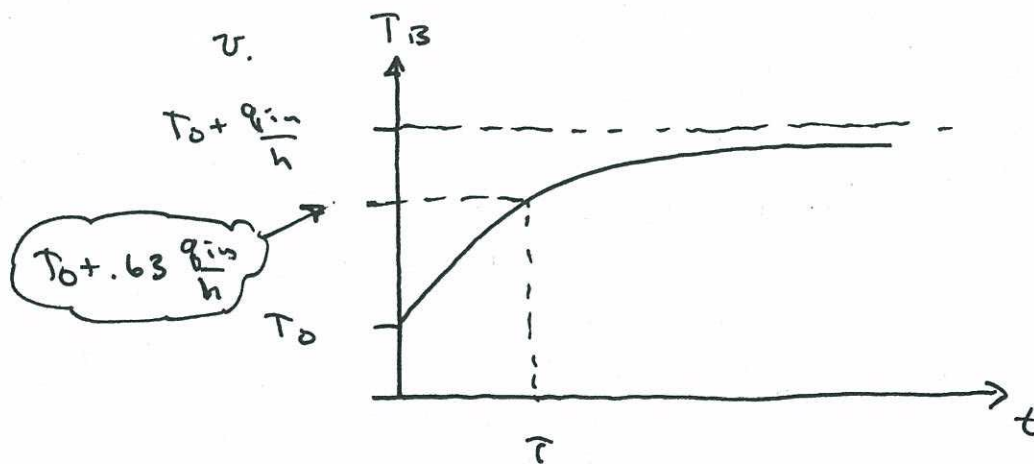
1(b) - continued

iv. Write equation in form $\frac{dy}{dt} + \frac{1}{\tau} y = f(t)$

$$\frac{d(T_B - T_0)}{dt} + \frac{h}{mC_p} (T_B - T_0) = \frac{\dot{q}_{in}}{mC_p}$$

$$\tau = \frac{mC_p}{h}$$

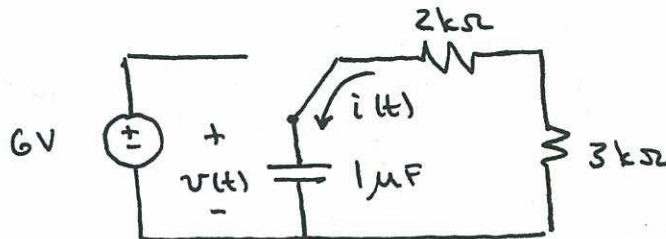
doubling mass doubles
time constant, as in
part (a)



Exercises
chapter 7.2

1.

(a) circuit for $t > 0$:



KVL, right loop: $v(t) + (3k\Omega)i(t) + (2k\Omega)i(t) = 0$

$$i(t) = C \frac{dv}{dt}$$

$$v(t) + 5k\Omega \left(1\mu F \frac{dv}{dt} \right) = 0$$

$$\underline{\underline{\frac{dv(t)}{dt} + 200 v(t) = 0}}$$

(b) $\tau = \frac{1}{200} \text{ sec} = \underline{\underline{0.005 \text{ sec}}}$

(c) Equivalent resistance:



$$R_{eq} = 5k\Omega$$

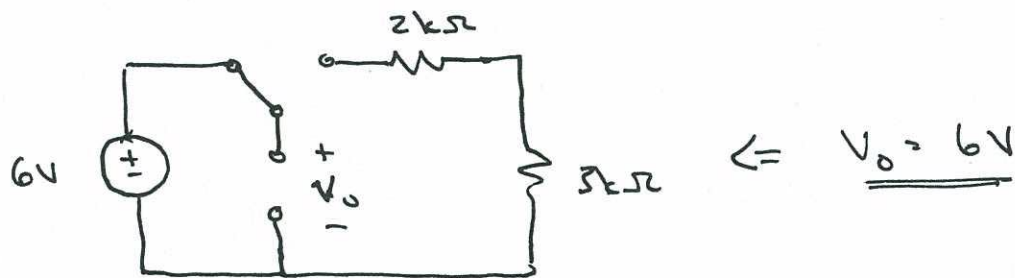
$$\tau = R_{eq} C = (5k\Omega)(1\mu F)$$

$$\underline{\underline{\tau = 0.005 \text{ sec}}}$$

CHECKS!

1. (cont'd)

(d) For $t < 0$, the switch is in position A and the capacitor looks like an open circuit (constant input, switch has been in position A for a long time) and the circuit looks like:



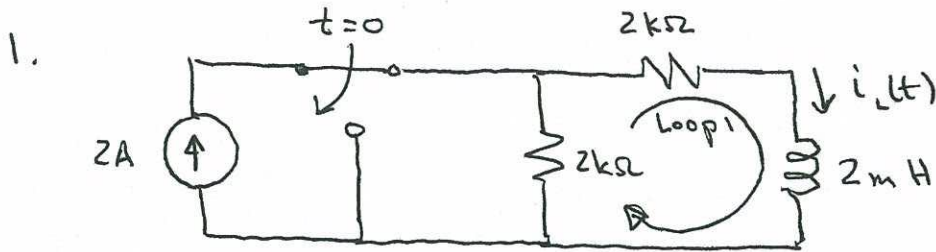
Voltages across capacitors can't change suddenly, so initial condition is:

$$v(t=0^+) = v(t=0^-) = \underline{6V}$$

(e) Form of solution is $v(t) = V_0 e^{-t/\tau}$, so:

$$\underline{v(t) = 6 e^{-200t} \text{ V}, t > 0}$$

Exercises
Chapter 7.3



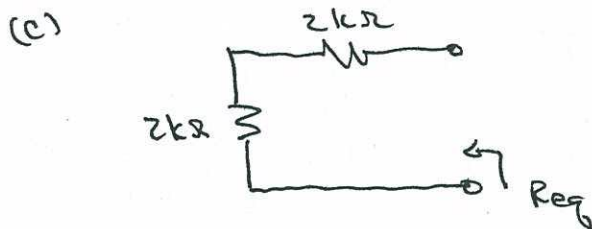
$$+ \quad L \frac{di_L}{dt}$$

(a) KVL, loop 1, $t > 0$:

$$(2k\Omega) i_L + (2k\Omega) i_L + 2mH \frac{di_L}{dt} = 0$$

$$\underline{\underline{\frac{di_L}{dt} + 2 \times 10^6 i_L = 0}}$$

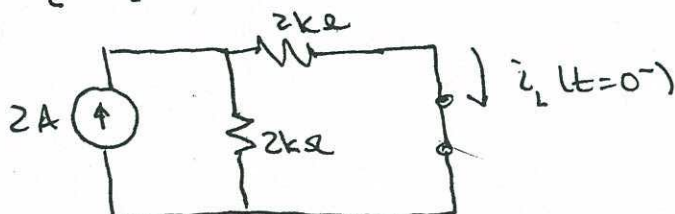
(b) From (a), $\tau = \frac{1}{2} \mu\text{sec}$



$$\Rightarrow R_{eq} = 2k\Omega + 2k\Omega = 4k\Omega$$

$$\tau = \frac{L}{R_{eq}} = \frac{2mH}{4k\Omega} = \underline{\underline{\frac{1}{2} \mu\text{sec}}}$$

(d) $t = 0^-$:



} inductor acts as short circuit at $t=0^-$

$$i_L(t=0^-) = 1A \quad \Leftarrow \text{circuit is a current divider}$$

$$i_L(t=0^+) = i_L(t=0^-) = \underline{\underline{1A}} \quad \Leftarrow \text{inductor current cannot change instantly.}$$

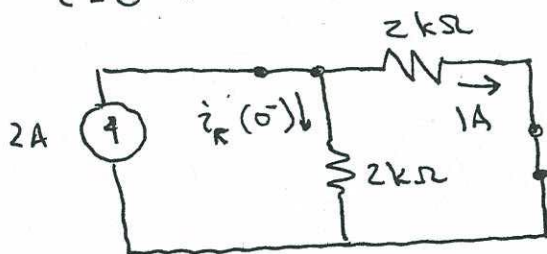
(e) From (a), $\frac{di_L}{dt} + 2 \times 10^6 i_L = 0 \Rightarrow \tau = 0.5$

From (b), $i_L(t=0^+) = 1\text{ A} = I_0$

Form of solution: $i_L(t) = I_0 e^{-t/\tau}$

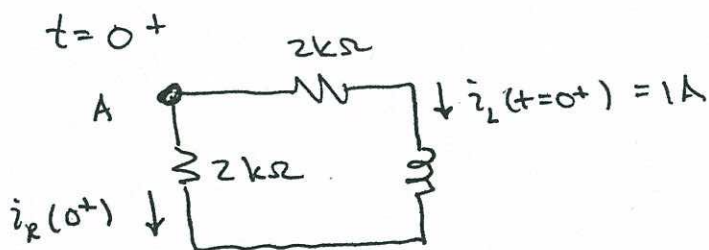
$i_L(t) = 1 e^{-2 \times 10^6 t}, t > 0$

(f) $t=0^-$:



$i_R(t=0^-) = 1\text{ A}$ (current divider)

$i_L(t=0^-) = 1\text{ A}$ (current divider)



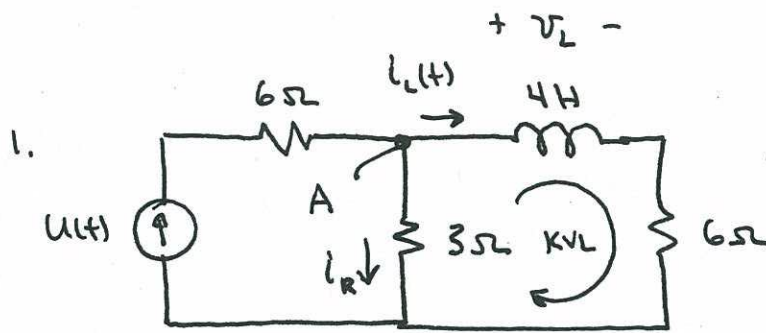
At $t=0^+$, $i_L(t=0^+) = 1\text{ A}$
(inductor current can't change suddenly)



KCL at A: $i_R(t=0^+) = -$

Resistor current is not continuous with time. (It changes suddenly to compensate for the change in applied current.) The other resistor's current is continuous with time.

Exercises
Chapter 7.4



KCL at node A: $i_R = u(t) - i_L(t)$

KVL around right loop:

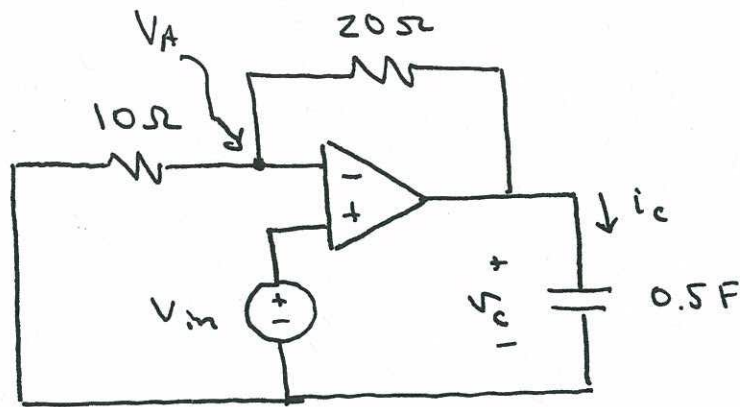
$$3\Omega(i_R) = v_L + 6\Omega(i_L) \quad ; \quad v_L = 4 \frac{di_L}{dt}$$

$$3\Omega(u - i_L) = 4 \frac{di_L}{dt} + 6\Omega(i_L)$$

$$\underline{\underline{3u = 4 \frac{di_L}{dt} + 9i_L}}$$

Exercises
Chapter 7.4

2.



Op-amp rules : $V_A = V_{in}$

KCL at inverting input : $\frac{V_A - 0V}{10\Omega} + \frac{V_A - V_c}{20\Omega} = 0$

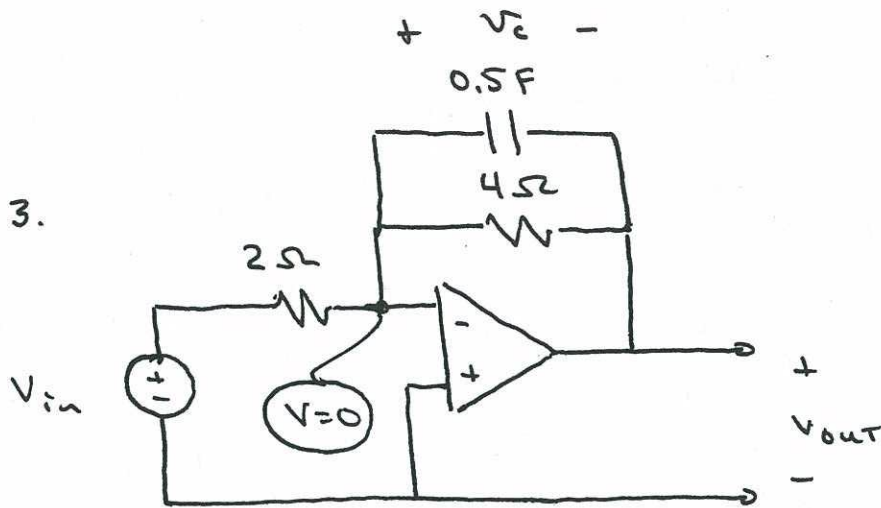
$$V_c = 3V_A = 3V_{in}$$

$$i_c = (0.5F) \frac{dV_c}{dt} = 0.5 \left[\frac{d}{dt} (3V_{in}) \right]$$

$$\underline{\underline{i_c(t) = \frac{3}{2} \frac{dV_{in}}{dt}}}$$

Exercises
Chapter 7.4

3.



KCL at inverting input:

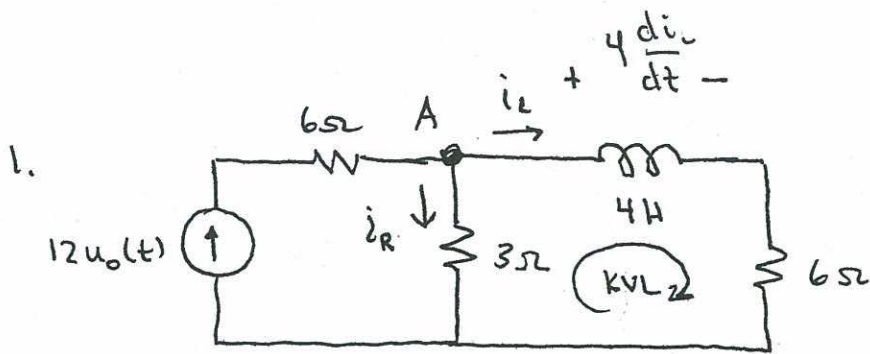
$$\frac{V_{in} - 0V}{2\Omega} = (0.5F) \frac{dV_c}{dt} + \frac{0 - V_{out}}{4\Omega}$$

$$V_c = -V_{out}$$

$$\frac{V_{in}}{2} = -0.5 \frac{dV_{out}}{dt} - \frac{V_{out}}{4}$$

$$\underline{\underline{2 \frac{dV_{out}}{dt} + V_{out} = -2V_{in}}}$$

Exercises
Chapter 7.5



(a) KCL at node A gives: $i_R = 12A - i_L$

KVL around right loop then results in:

$$3\Omega (12A - i_L) = 4 \frac{di_L}{dt} + 6\Omega (i_L)$$

$$36 = 4 \frac{di_L}{dt} + 6i_L$$

$$\underline{\underline{\frac{di_L}{dt} + \frac{9}{4} i_L = 9}}$$

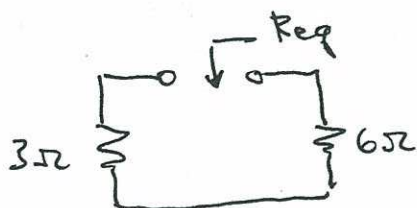
(b) From (a), $\frac{9}{4} = \frac{1}{\tau} \Rightarrow \underline{\underline{\tau = \frac{4}{9} \text{ sec}}}$

(c) $i_L(t) = K_1 + K_2 e^{-t/\tau}$

(d) $t = 0^+ \Rightarrow i_L(0^+) = i_L(0^-) = 0A$
 $t \rightarrow \infty \Rightarrow i_L(t \rightarrow \infty) = 4A$

$\left. \begin{array}{l} t = 0^+ \Rightarrow i_L(0^+) = i_L(0^-) = 0A \\ t \rightarrow \infty \Rightarrow i_L(t \rightarrow \infty) = 4A \end{array} \right\} \Rightarrow \underline{\underline{i_L(t) = 4[1 - e^{-4t/9}]}}$
 $(t > 0)$

(e)



$$\Rightarrow R_{eq} = 3\Omega + 6\Omega = 9\Omega$$

$$\tau = \frac{L}{R} = \frac{4H}{9\Omega} \Rightarrow \underline{\underline{\tau = \frac{4}{9} \text{ sec} \checkmark}}$$

Exercises
chapter 7.5

2. $2 \frac{di(t)}{dt} + 3i(t) = 5v(t)$

DC gain: $\frac{di}{dt} \rightarrow 0 \Rightarrow 3i(t \rightarrow \infty) = 5v(t \rightarrow \infty)$

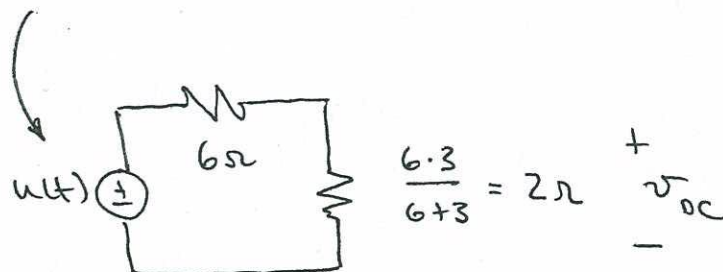
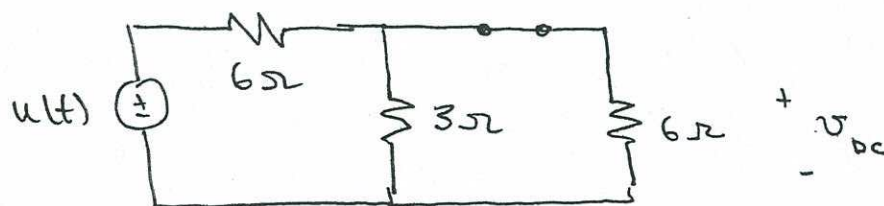
$$\left. \frac{i(t)}{v(t)} \right|_{t \rightarrow \infty} = \underline{\underline{\frac{5}{3}}}$$

Units are $\frac{A}{V}$.

Time constant: $\frac{di(t)}{dt} + \frac{3}{2}i(t) = \frac{5}{2}v(t)$

$\tau = \underline{\underline{\frac{2}{3} \text{ sec}}}$

3. In steady-state, with $u(t) = \text{constant}$, the inductors becomes a short circuit:



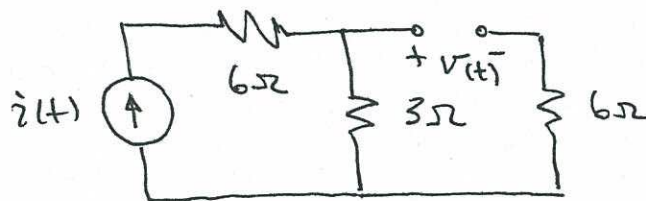
Voltage divider: $v_{oc} = \frac{2}{8} u \Rightarrow \text{DC gain} = \underline{\underline{\frac{1}{4}}}$
units are $\frac{V}{V}$
(unitless)

Exercises
Chapter 7.5

4. From differential equation, with $\frac{dv}{dt} = 0$:

$$\frac{1}{9} v(t \rightarrow \infty) = i(t \rightarrow \infty) \Rightarrow \text{DC gain} = 9 \quad (\text{from Diff. eqn.})$$

From circuit, with capacitor open-circuited:



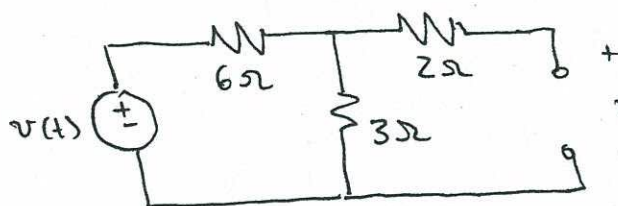
$$v(t) = 3 i(t)$$

$$\Rightarrow \text{D gain} = 3$$

DC gain from circuit different than DC gain from differential equation $\Rightarrow$ D.E. not accurate

5. $3 \frac{dv_c}{dt} + v_c(t) = \dot{v}(t) \Rightarrow \text{DC gain} = 1$

circuit:



$$\Rightarrow \text{DC gain} = \frac{1}{2}$$

Inconsistent

$\Rightarrow$ Diff. equation probably inaccurate.

Exercises
Chapter 8.1

1. $\frac{d^2 y(t)}{dt^2} + 32 \frac{dy(t)}{dt} + 64 y(t) = 0$

(a) $\omega_n = \sqrt{64} = \underline{\underline{8 \text{ rad/sec}}}$

(b) $2 \zeta \omega_n = 32 \Rightarrow 2 \zeta (8) = 32 \Rightarrow \underline{\underline{\zeta = 2}}$

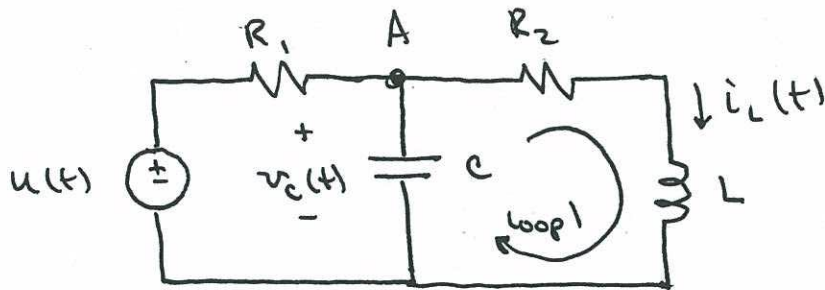
2. $\frac{d^2 y(t)}{dt^2} + 8 \frac{dy(t)}{dt} + 64 y(t) = 0$

(a) $\omega_n = \sqrt{64} = 8 \text{ rad/sec}$

(b) $2 \zeta \omega_n = 8 \Rightarrow 2 \zeta (8) = 8 \Rightarrow \underline{\underline{\zeta = 0.5}}$

Exercises
chapter 8.1

3.



$$\text{KCL at A: } \frac{u - v_c}{R_1} = C \frac{dv_c}{dt} + i_L \quad (1)$$

$$\text{KVL, Loop 1: } v_c = R_2 i_L + L \frac{di_L}{dt} \quad (2)$$

$$\text{From (1), } i_L = \frac{u}{R_1} - \frac{v_c}{R_1} - C \frac{dv_c}{dt} \quad (3)$$

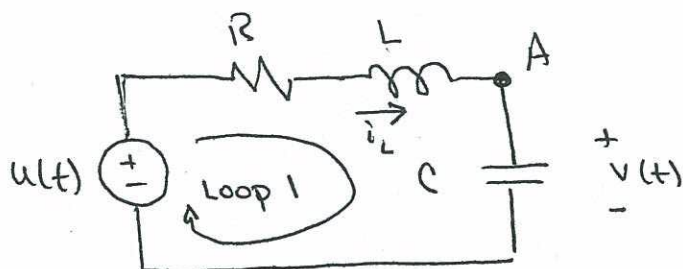
$$\frac{di_L}{dt} = \frac{1}{R_1} \frac{du}{dt} - \frac{1}{R_1} \frac{dv_c}{dt} - C \frac{d^2 v_c}{dt^2} \quad (4)$$

Substitute (3) & (4) into (2)

$$v_c = R_2 \left[\frac{u}{R_1} - \frac{v_c}{R_1} - C \frac{dv_c}{dt} \right] + L \left[\frac{1}{R_1} \frac{du}{dt} - \frac{1}{R_1} \frac{dv_c}{dt} - C \frac{d^2 v_c}{dt^2} \right]$$

↖
This can be simplified, but that's
not required by the problem
statement...

4.



Exercises
chapter 8.1

$$\text{KCL at A: } \underline{i_L = C \frac{dv}{dt}}$$

$$\text{KVL: } \underline{u(t) = Ri_L + L \frac{di_L}{dt} + v}$$

$$i_L = C \frac{dv}{dt} \Rightarrow \frac{di_L}{dt} = C \frac{d^2v}{dt^2}$$

$$u(t) = Ri_L + L \frac{di_L}{dt} + v$$

$$\underline{\underline{u(t) = RC \frac{dv}{dt} + LC \frac{d^2v}{dt^2} + v}}$$

Exercises

Chapter 8.3

1. $\cos(\pi t - 30^\circ)$

2. $v(t) = \text{Real} \{ e^{j100t} \}$

3. $v = 3 - j3 = \sqrt{3^2 + 3^2} \angle \tan^{-1}(-\frac{3}{3}) = \underline{3\sqrt{2} \angle -45^\circ}$

or equivalently:
 $3\sqrt{2} e^{-j45^\circ}$

4. $e^{j\pi} = 1 e^{j180^\circ} = \underline{\underline{-1}}$

5. $e^{-j\pi/2} = \underline{\underline{-j}}$

Exercises
Chapter 8.4

$$1. \quad \frac{d^2 y}{dt^2} + 6 \frac{dy}{dt} + 144 y = 0$$

$\uparrow$ $\uparrow$
 $2\zeta\omega_n$ ω_n^2

$$\omega_n^2 = 144 \Rightarrow \omega_n = \underline{\underline{12 \text{ rad/sec}}}$$

$$2\zeta\omega_n = 6 \Rightarrow 2\zeta(12) = 6 \Rightarrow \underline{\underline{\zeta = \frac{1}{4}}}$$

circuit is underdamped ($\zeta < 1$)

$$2. \quad \frac{d^2 y}{dt^2} + 32 \frac{dy}{dt} + 64 y = 0$$

$$(a) \quad \omega_n = \sqrt{64} \Rightarrow \omega_n = \underline{\underline{8 \text{ rad/sec}}}$$

$$(b) \quad 2\zeta\omega_n = 32 \Rightarrow 2\zeta(8) = 32 \Rightarrow \underline{\underline{\zeta = 2}}$$

$$(c) \quad -\frac{1}{\tau_{1,2}} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

$$\tau_1 = + \left[\frac{1}{\zeta\omega_n + \omega_n \sqrt{\zeta^2 - 1}} \right] = \frac{1}{2(8) + 8} = \underline{\underline{\frac{1}{24} \text{ sec}}}$$

$$\tau_2 = + \left[\frac{1}{\zeta\omega_n - \omega_n \sqrt{\zeta^2 - 1}} \right] = \frac{1}{2(8) - 8} = \underline{\underline{\frac{1}{8} \text{ sec}}}$$

Exercises

chapter 8.4

$$3. \quad \frac{d^2 y}{dt^2} + 8 \frac{dy}{dt} + 64y = 0$$

$$(a) \quad \omega_n = \sqrt{64} = \underline{\underline{8 \text{ rad/s}}}$$

$$(b) \quad 2\zeta\omega_n = 8 \Rightarrow 2\zeta(8) = 8 \Rightarrow \underline{\underline{\zeta = \frac{1}{2}}}$$

$$(c) \quad \omega_d = \omega_n \sqrt{1 - \zeta^2} = 8 \sqrt{1 - .25}$$

$$\underline{\underline{\omega_d = 6.93 \text{ rad/sec}}}$$

Exercises
chapter 8.5

$$1. \quad \frac{d^2 v}{dt^2} + 12 \frac{dv}{dt} + 400v = 400u(t)$$

$$u(t) = 2u_0(t)$$

$$(a) \quad \omega_n^2 = 400 \Rightarrow \underline{\omega_n = 20 \text{ rad/s}}$$

$$(b) \quad 2\zeta\omega_n = 12 \Rightarrow 2\zeta(20) = 12 \Rightarrow \underline{\zeta = 0.3}$$

$$(c) \quad M_p \approx 1 - \frac{\zeta}{0.6} = \underline{0.5}$$

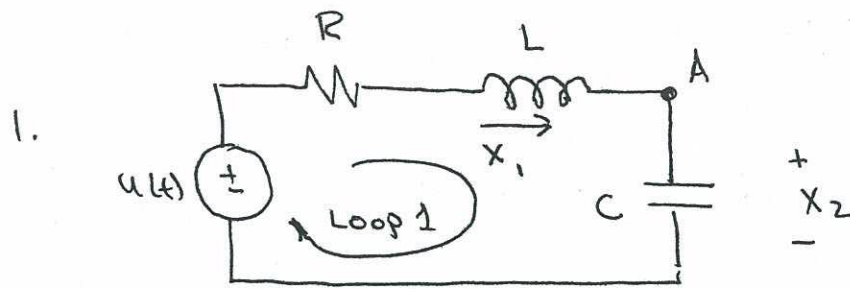
$$t_r = \frac{1.8}{\omega_n} = \frac{1.8}{20} \Rightarrow \underline{t_r = 0.09 \text{ sec}}$$

$$400v_{ss} = 400[2u_0(t \rightarrow \infty)] \quad \leftarrow \frac{dv}{dt}, \frac{d^2v}{dt^2} = 0$$

$$\underline{v_{ss} = 2V}$$

$$(d) \quad v_{\max} = v_{ss} [1 + M_p] \Rightarrow v_{\max} = 2[1 + 0.5] = \underline{3V}$$

Exercises
chapter 9.1



States : x_1 = current through inductor

x_2 = voltage across capacitor

$$\text{KCL: } x_1 = C \frac{dx_2}{dt} = C \dot{x}_2 \Rightarrow \dot{x}_2 = \frac{1}{C} x_1$$

$$\text{KVL: } u = L \frac{dx_1}{dt} + R x_1 + x_2 \Rightarrow \dot{x}_1 = \frac{u}{L} - \frac{R}{L} x_1 - \frac{1}{L} x_2$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix}}_{\underline{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}}_{\underline{b}} u$$

2. $\{A, \underline{b}\}$ as in problem 1.

output equation: $y = x_2 \Rightarrow c = [0 \ 1], d = 0$

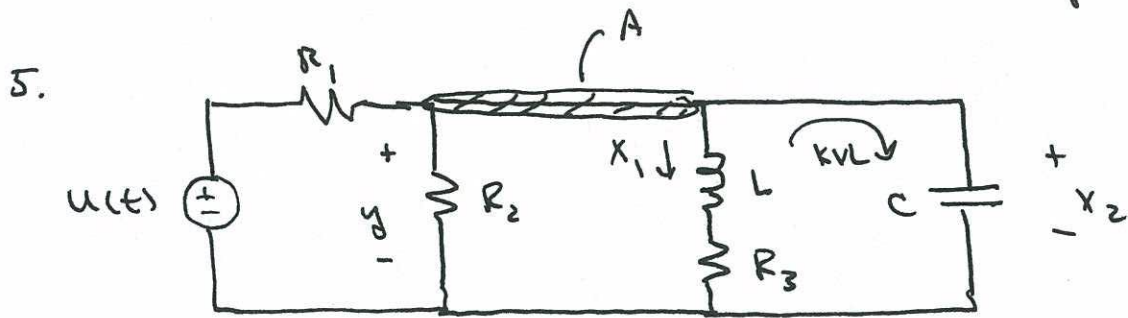
3. $\{A, \underline{b}\}$ as in problem 1

output equation: $y = u - R x_1 - x_2 \Rightarrow c = [-R \ -1], d = 1$

4. $\{A, \underline{b}\}$ as in problem 1

output equation: $y = R x_1 \Rightarrow c = [R \ 0], d = 0$

Exercises
Chapter 9.1



KCL at node A:

$$\frac{u - x_2}{R_1} = \frac{x_2}{R_2} + \dot{x}_1 + C \dot{x}_2 \Rightarrow \dot{x}_2 = -\frac{1}{C} x_1 - \frac{1}{C} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) x_2 + \frac{u}{R_1 C}$$

KVL, right loop:

$$L \ddot{x}_1 + R_3 \dot{x}_1 = x_2 \Rightarrow \ddot{x}_1 = -\frac{1}{R_3 L} x_1 + \frac{1}{L} x_2$$

Output equation: $y = x_2$

In matrix form, the model is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_3 L} & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{C} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{R_1 C} \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 0 \cdot u$$

Exercises

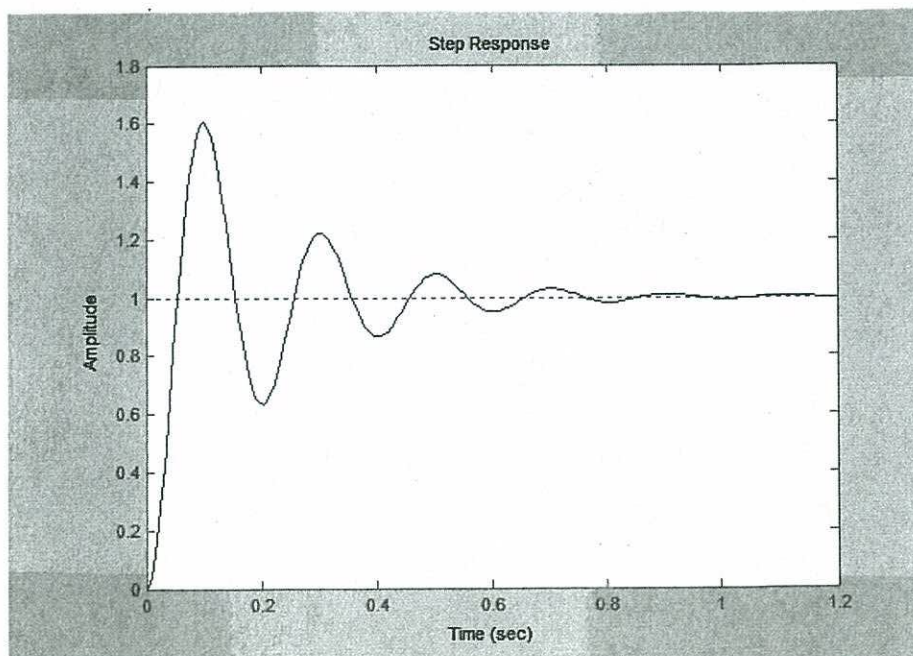
Chapter 9.2

1. State variable model for this circuit was created in exercise 1 of chapter 9.1.

We just need to plug in numbers and use MATLAB's "step" command. The commands I used are below:

```
R = 10;  
L = 1;  
C = 1e-3;  
  
A = [-R/L -1/L; 1/C 0];  
b = [1/L; 0];  
c = [0 1];  
d = 0;  
  
sys = ss(A,b,c,d);  
step(sys)
```

Resulting Figure:

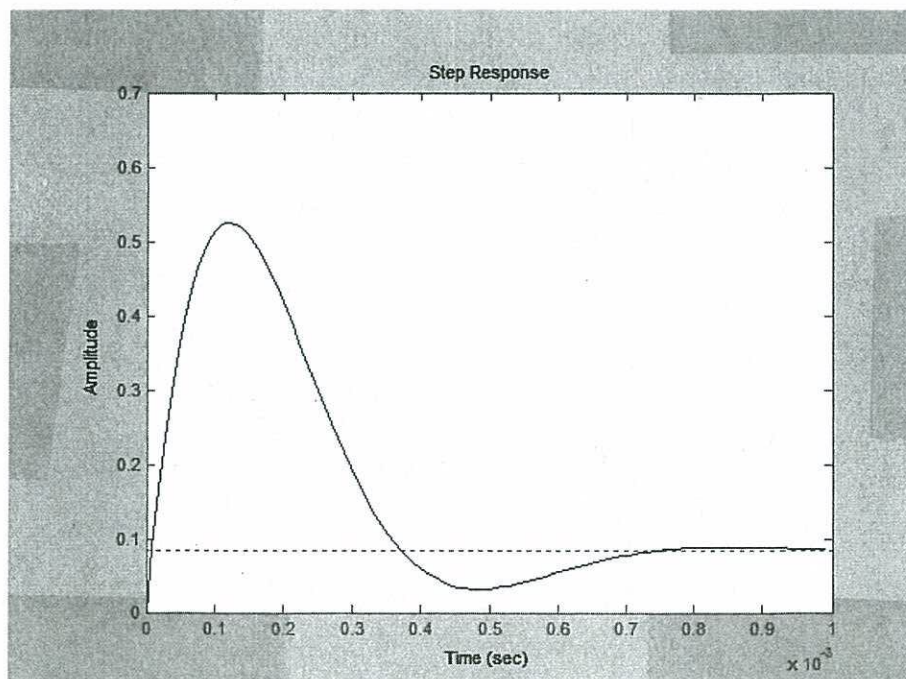


Exercises Chapter 9.2

2. The state variable model for this circuit was created in exercise 2 of chapter 9.1. We just need to plug in numbers and use MATLAB's "step" command, as shown below:

```
R1 = 10;  
R2 = 100;  
R3 = 1.1;  
L = 1e-3;  
C = 10e-6;  
  
A = [-1/R3/L 1/L; -1/C -1/C*(1/R1+1/R2)];  
b = [0; 1/(R1*C)];  
c = [0 1];  
d = 0;  
  
sys = ss(A,b,c,d);  
step(sys)
```

Resulting Figure:



Exercises Chapter 9.3

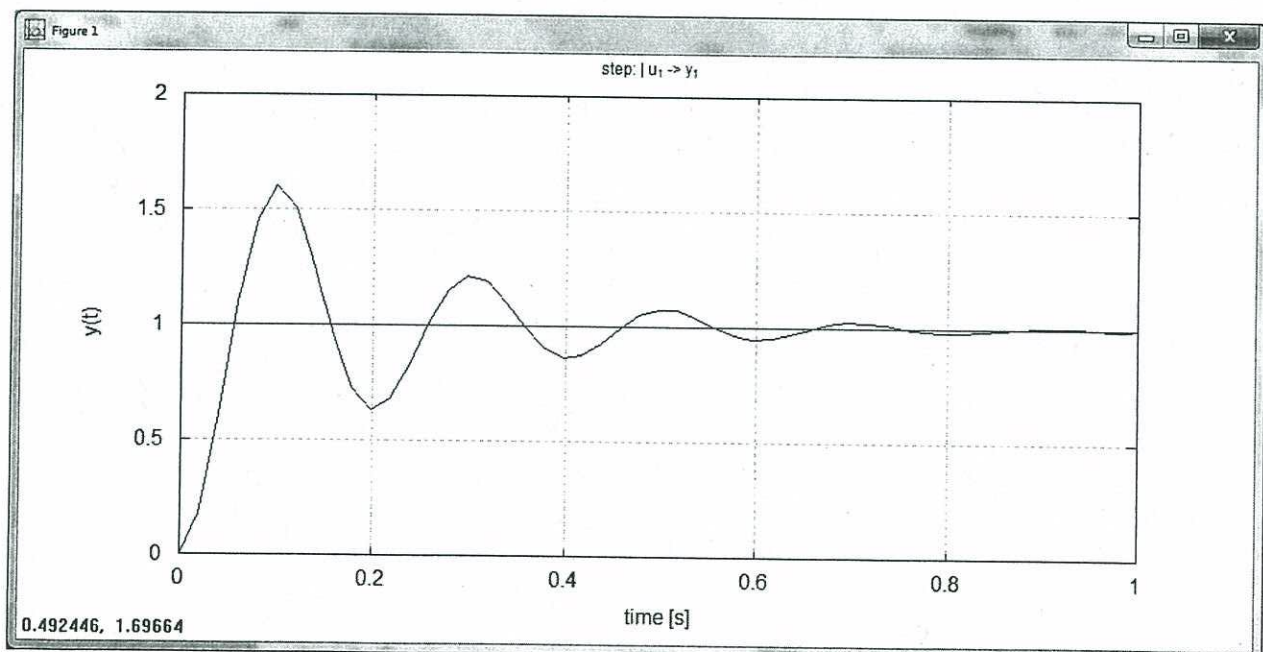
1. The state variable model for this circuit was created in exercise 1 of chapter 9.1. We can just plug in numbers and do the stimulation with Octave:

Octave commands:

```
R = 10;  
L = 1;  
C = 1e-3;  
  
A = [-R/L -1/L; 1/C 0];  
b = [1/L; 0];  
c = [0 1];  
d = 0;
```

```
sys = ss(A,b,c,d);  
step(sys)
```

Resulting Figure:



Exercises chapter 9.3

2. The state variable model for this circuit was created in exercise 2 of chapter 9.1. We just need to plug in numbers and use Octave to simulate the response:

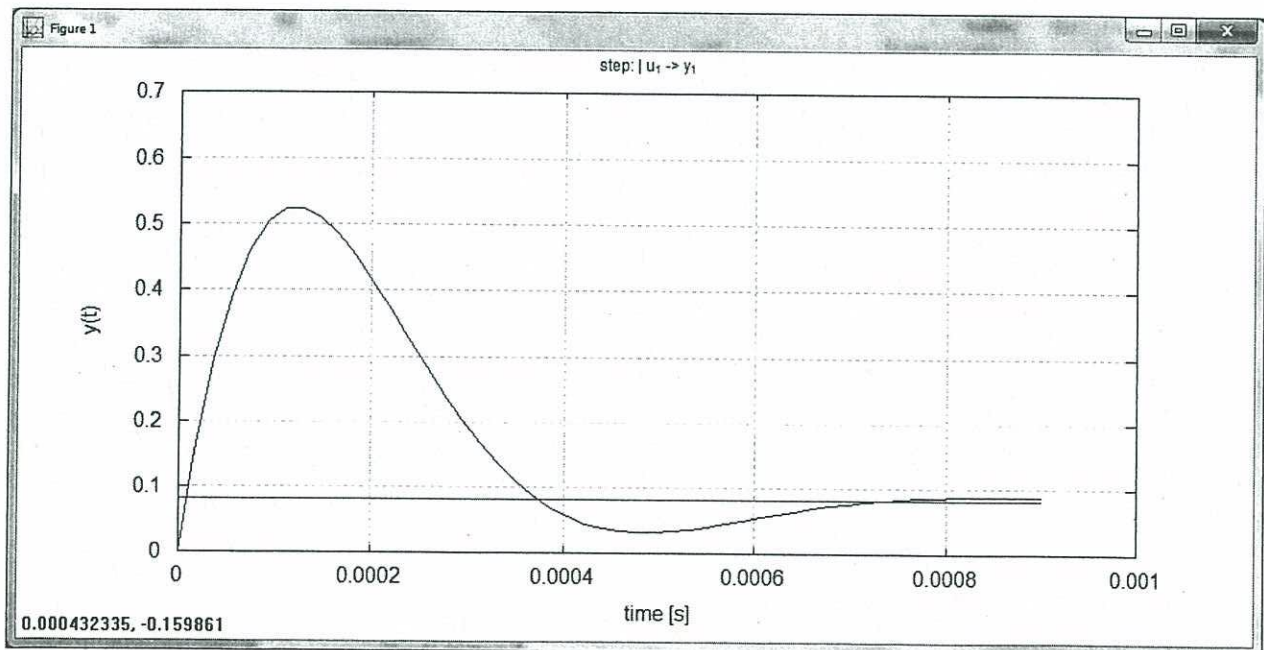
Octave commands:

```
R1 = 10;  
R2 = 100;  
R3 = 1.1;  
L = 1e-3;  
C = 10e-6;
```

```
A = [-1/R3/L 1/L; -1/C -1/C*(1/R1+1/R2)];  
b = [0; 1/(R1*C)];  
c = [0 1];  
d = 0;
```

```
sys = ss(A,b,c,d);  
step(sys)
```

Resulting figure:



Exercises
Chapter 10.1

1. $V_{in}(t) = 10 \cos(2t + 40^\circ)$

Input frequency = 2 rad/sec , so output
frequency must be 2 rad/sec

2. $V_{in}(t) = 10 \cos(2t + 40^\circ)$

$$\left| \frac{V_{out}}{V_{in}} \right| = 0.5, \quad \text{Input phase} - \text{output phase} = 20^\circ$$

(a) 2 rad/sec (same as input)

(b) $\left| \frac{V_{out}}{V_{in}} \right| = 0.5 \Rightarrow |V_{out}| = 0.5 |V_{in}| = 0.5(10)$

$$|V_{out}| = 5 \text{ V}$$

(c) $40^\circ - \phi = 20^\circ \Rightarrow \phi = 20^\circ$

↑
input phase

Note: Output wave form is, thus:

$$V_{out}(t) = 5 \cos(2t + 20^\circ)$$

chapter 10.2

Exercises

1.

$$(a) \frac{3}{\sqrt{2}} (1+j)$$

$$(b) 5(-1+j)$$

$$(c) j2.5$$

$$(d) -6$$

2.

$$(a) 2\sqrt{2} e^{-j45^\circ}$$

$$(b) 3 e^{-j90^\circ}$$

$$(c) 6$$

$$(d) \sqrt{10} e^{j18.4^\circ}$$

3.

$$(a) \frac{-j}{2(j+1)} = \frac{1 \angle -90^\circ}{2\sqrt{2} \angle 45^\circ} = \frac{1}{2\sqrt{2}} \angle -135^\circ = \underline{\underline{\frac{1}{2\sqrt{2}} e^{j(-135^\circ)}}}$$

$$(b) \frac{2-j2}{4+j4} = \frac{2\sqrt{2} \angle -45^\circ}{4\sqrt{2} \angle +45^\circ} = \underline{\underline{\frac{1}{2} \angle -90^\circ}} = \underline{\underline{\frac{1}{2} e^{j(-90^\circ)}}} = -\underline{\underline{\frac{j}{2}}}$$

$$(c) 2 e^{j45^\circ} \frac{2}{j+1} = 2 \angle 45^\circ \frac{2}{\sqrt{2} \angle 45^\circ} = \underline{\underline{\frac{4}{\sqrt{2}}}} = \underline{\underline{2\sqrt{2}}}$$

$$(d) j + \frac{2}{j} = j + \frac{2}{1 \angle 90^\circ} = j + 2 \angle -90^\circ = j - j2 = -j = \underline{\underline{e^{-j90^\circ}}}$$

Chapter 10.2
Exercises

4. (a) $3 \angle -60^\circ$ or $3e^{-j60^\circ}$
(b) $-2 \angle 45^\circ = 2 \angle 225^\circ$ or $2e^{j225^\circ}$
(c) $\sin(6t) = \cos(6t - 90^\circ) \Rightarrow 1 \angle -90^\circ$ or $1e^{-j90^\circ}$
(d) $7 \angle 0^\circ = 7$ or $7e^{j0^\circ}$

5. (a) $5\sqrt{2} \cos(100t - 45^\circ)$
(b) $3 \cos(3t + \pi)$
(c) $2 \cos(\pi t - 30^\circ) + 4 \cos(4t + 20^\circ)$

Exercises
chapter 10.3

1.

$$u(t) = 2 \cos(3t)$$

$$\text{let } u(t) = 2e^{j3t}, \quad y(t) = \underline{Y} e^{j3t} \Rightarrow \frac{dy}{dt} = j3 \underline{Y} e^{j3t}$$

$$2 \frac{dy}{dt} + 6y(t) = u(t) \Rightarrow 2[j3 \underline{Y} e^{j3t}] + 6[\underline{Y} e^{j3t}] = 2e^{j3t}$$

$$j6 \underline{Y} e^{j3t} + 6 \underline{Y} e^{j3t} = 2e^{j3t}$$

$$\underline{Y} = \frac{2}{6 + j6} = \frac{2}{6\sqrt{2} \angle 45^\circ} = \frac{1}{3\sqrt{2}} \angle -45^\circ$$

$$\underline{y(t)} = \underline{\frac{1}{3\sqrt{2}} \cos(3t - 45^\circ)}$$

2.



KCL at A:

$$\frac{u - y}{1\Omega} = 0.5F \frac{dy}{dt}$$

$$\frac{dy}{dt} + 2y = 2u$$

(b) $u(t) = 3 \cos(2t) \Rightarrow \text{let } u(t) = 3e^{j2t} \quad \& \quad y(t) = \underline{Y} e^{j2t}$

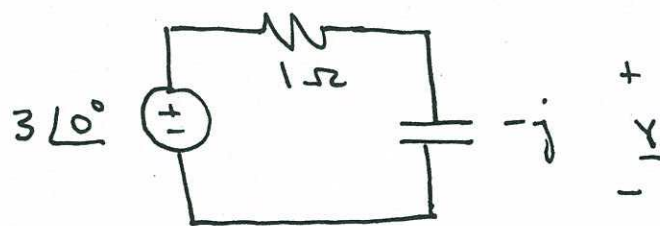
$$(j2) \underline{Y} e^{j2t} + 2 \underline{Y} e^{j2t} = 2(3e^{j2t})$$

$$\underline{Y} = \frac{6}{2 + j2} = \frac{6}{2\sqrt{2} \angle 45^\circ} = \frac{3}{\sqrt{2}} \angle -45^\circ \Rightarrow \underline{y(t)} = \underline{\underline{\frac{3}{\sqrt{2}} \cos(2t - 45^\circ)}}$$

Exercises

Chapter 10.4.

1. We did this problem in exercise 2 of chapter 10.3, but let's do it differently here. (This approach won't be formally introduced until chapter 10.5)
 - At a frequency of 2 rad/sec, the capacitor impedance is $\frac{1}{j(2)(0.5)} = -j$.
 - Substitute the frequency-domain representations of the input and the capacitor into the circuit:



Now use voltage divider formula to find

$\underline{Y}$:

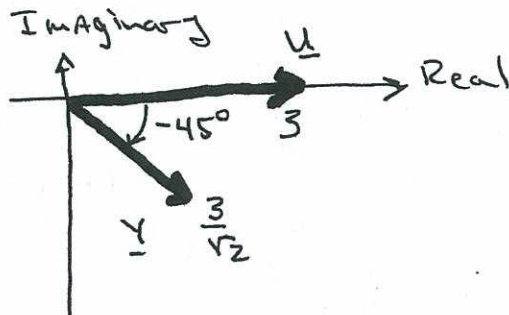
$$\underline{Y} = 3\angle 0^\circ \left(\frac{-j}{1-j} \right) = 3\angle 0^\circ \left(\frac{1\angle -90^\circ}{\sqrt{2}\angle -45^\circ} \right)$$

$$\underline{Y} = \frac{3}{\sqrt{2}} \angle -45^\circ \Rightarrow \underline{y(t)} = \underline{\underline{\frac{3}{\sqrt{2}} \cos(2t - 45^\circ)}}$$

Exercises
Chapter 10.4

2.

$$\underline{U} = 3 \angle 0^\circ, \quad \underline{Y} = \frac{3}{\sqrt{2}} \angle -45^\circ$$



3.

$$2 \cos 4t \text{ across } \frac{1}{8} \text{ F} \Rightarrow Z = \frac{1}{j\omega C} = \frac{1}{j(4)\frac{1}{8}} = \underline{\underline{-j2\Omega}}$$

$$2 \cos 4t \text{ across } 2 \times 10^{-3} \text{ H} \Rightarrow Z = j\omega L = j(4)(2 \times 10^{-3}) = \underline{\underline{j8 \text{ m}\Omega}}$$

$$5 \text{ k}\Omega \Rightarrow \underline{\underline{Z = 5 \text{ k}\Omega}}$$

4.

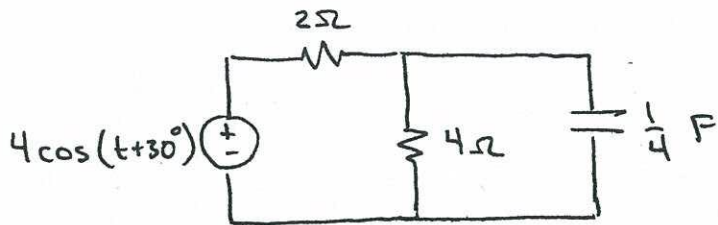
$$3 \cos 8t \text{ across } \frac{1}{8} \text{ F} \Rightarrow Z = \frac{1}{j\omega C} = \frac{1}{j(8)\frac{1}{8}} = \underline{\underline{-j1\Omega}}$$

$$3 \cos 8t \text{ across } 2 \text{ mH} \Rightarrow Z = j\omega L = j(8)(2 \times 10^{-3}) = j16 \text{ m}\Omega$$

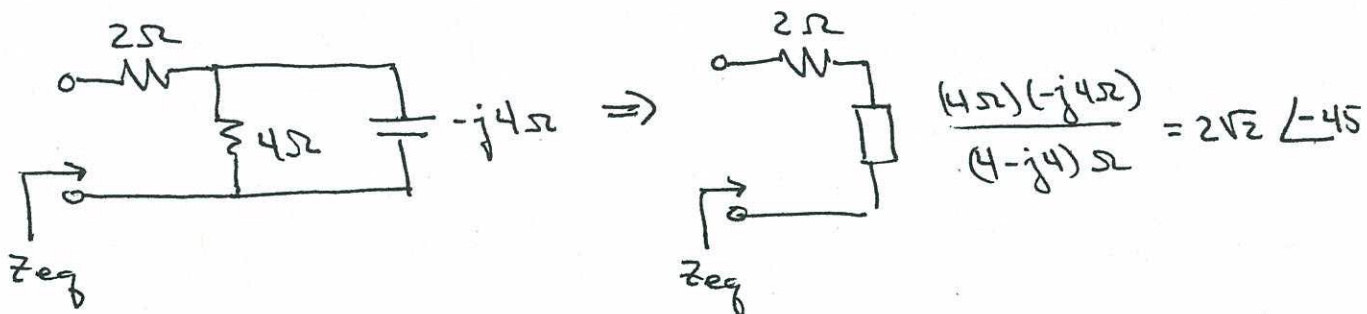
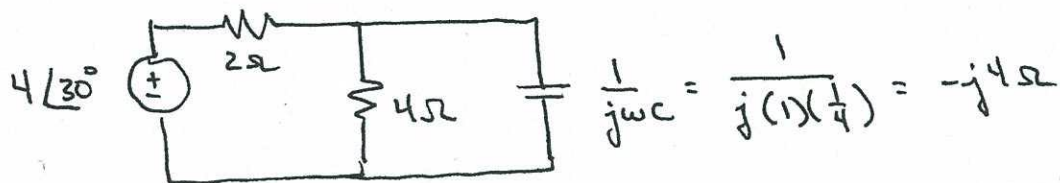
5 k Ω resistor's impedance is 5 k Ω , independent of frequency

Exercises
Chapter 10.5

1.

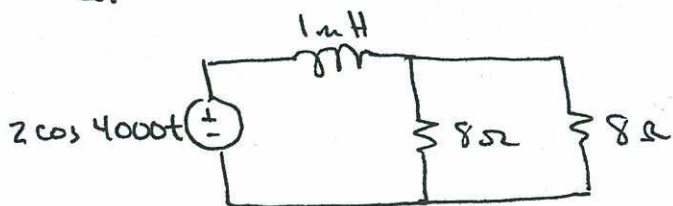


$\omega = 1 \text{ rad/sec}$

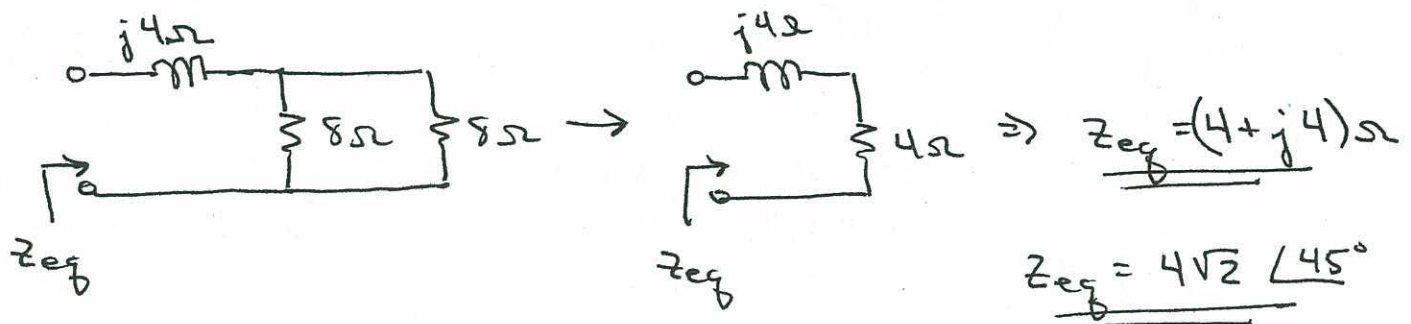
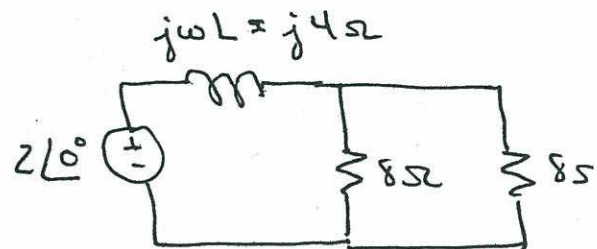


$$Z_{eq} = 2\Omega + 2\sqrt{2} \angle -45^\circ = 2 + (2 - j2) = \underline{4 - j2\Omega} = \underline{2\sqrt{5} \angle -26.6^\circ}$$

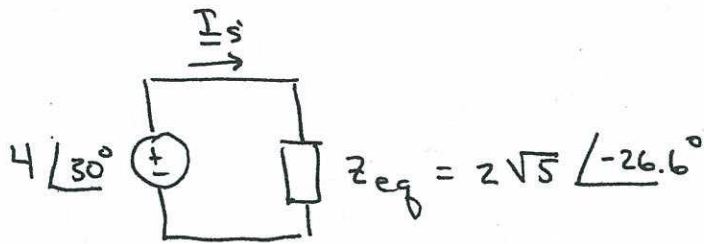
2.



$\omega = 4000 \text{ rad/s}$



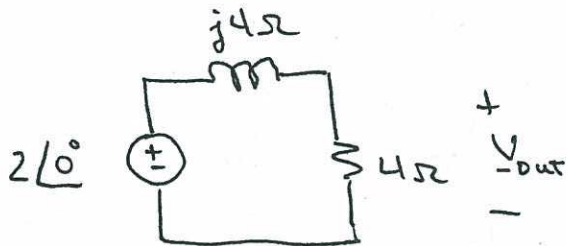
3. From exercise 1, the equivalent circuit is



$$\underline{I}_s = \frac{4 \angle 30^\circ}{2\sqrt{5} \angle -26.6^\circ} = \frac{2}{\sqrt{5}} \angle 56.6^\circ$$

$$\underline{i}_s(t) = \underline{\underline{\frac{2}{\sqrt{5}} \cos(t + 56.6^\circ)}}$$

4. From exercise 2, the equivalent circuit is:



$$\underline{V}_{out} = 2 \angle 0^\circ \left[\frac{4\Omega}{(4 + j4)\Omega} \right] = 2 \angle 0^\circ \left[\frac{4}{4\sqrt{2} \angle 45^\circ} \right] = \frac{8}{4\sqrt{2}} \angle -45^\circ$$

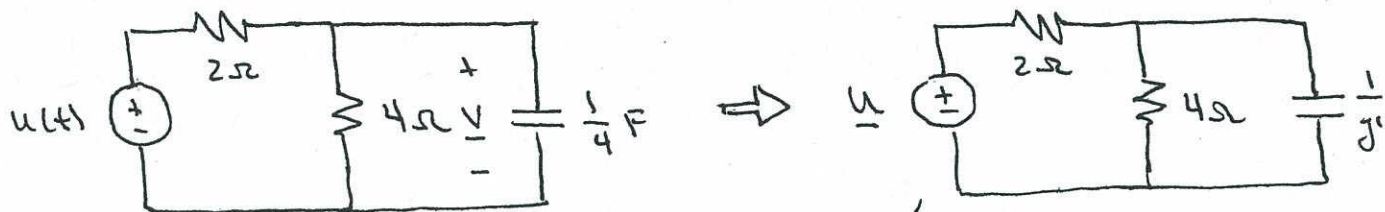
$$\underline{V}_{out} = \sqrt{2} \angle -45^\circ$$

$$\underline{\underline{v_{out}(t) = \sqrt{2} \cos(4000t - 45^\circ)}}$$

Exercises
chapter 10.6

1. $u(t) = 4 \cos(t + 30^\circ) + 2 \cos(2t - 45^\circ)$

$\uparrow$ $\uparrow$
 $\omega = 1$ $\omega = 2$



circuit reduction

$$\frac{4 \cdot \frac{1}{j\omega C}}{4 + \frac{1}{j\omega C}} = \frac{4}{4j\omega C + 1} = \frac{4}{j4\omega(\frac{1}{4}) + 1} = \frac{4}{j\omega + 1}$$

voltage divider:

$$\underline{V} = \underline{U} \frac{\frac{4}{j\omega + 1}}{2 + \frac{4}{j\omega + 1}} = \frac{4}{6 + j2\omega} \underline{U}$$

$$\omega = 1 \text{ rad/sec} \Rightarrow \underline{V} = \frac{4}{6 + j2(1)} [4 \angle 30^\circ] = (0.63 \angle -18.4^\circ) (4 \angle 30^\circ)$$

$$\underline{V} = 2.52 \angle 11.6^\circ$$

$$\omega = 2 \text{ rad/sec} \Rightarrow \underline{V} = \frac{4}{6 + j2(2)} [2 \angle -45^\circ] = 1.1 \angle -78.7^\circ$$

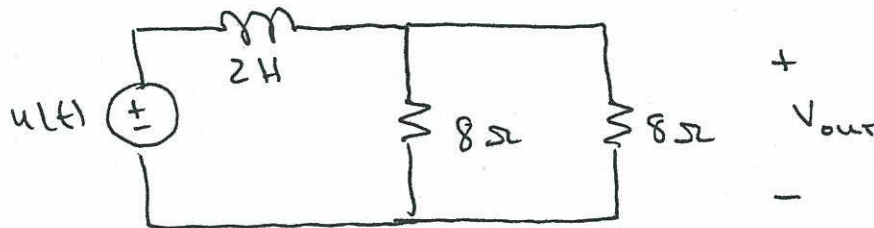
superimpose in time domain:

$$\underline{v(t)} = \underline{2.52 \cos(t + 11.6^\circ) + 1.1 \cos(2t - 78.7^\circ)}$$

Exercises
Chapter 10.6

2. Total applied voltage:

$$u(t) = u_1(t) + u_2(t) = 4 \cos(2t) + \cos(4t)$$



Circuit reduction & conversion to frequency domain



$$\underline{V}_{out} = \frac{4}{4 + j2\omega} \cdot \underline{u} \quad \leftarrow \text{Voltage divider}$$

$$\omega = 2 \text{ rad/sec} \Rightarrow \underline{V}_{out} = \frac{4}{4 + j4} (4 \angle 0^\circ) = 2\sqrt{2} \angle -45^\circ$$

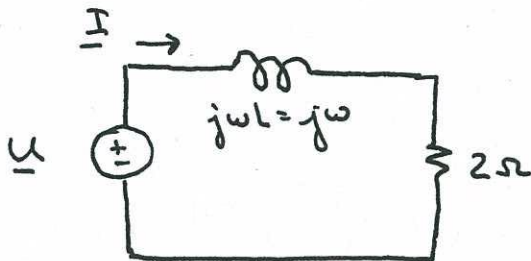
$$\omega = 4 \text{ rad/sec} \Rightarrow \underline{V}_{out} = \frac{4}{4 + j8} (1 \angle 0^\circ) = 0.45 \angle -63.4^\circ$$

Superimpose in time domain to get total response:

$$\underline{\underline{v_{out}(t) = 2\sqrt{2} \cos(2t - 45^\circ) + 0.45 \cos(4t - 63.4^\circ)}}$$

Exercises
chapter 11.1

1. Frequency domain model:



$$\underline{I} = \frac{\underline{u}}{2 + j\omega} \Rightarrow \underline{\underline{\frac{I}{u} = \frac{1}{2 + j\omega}}}$$

2. $u(t) = 3 \cos(2t - 60^\circ) + 4 \cos(4t + 30^\circ) + 7 \cos(6t + 45^\circ)$

• $\omega = 2 \text{ rad/s} \Rightarrow \underline{u}_1 = 3 \angle -60^\circ, |H(j2)| = 0, \angle H(j2) = 0^\circ$

$$\underline{y}_1 = \underline{u}_1 (0 \angle 0^\circ) = 0 \angle -60^\circ$$

• $\omega = 4 \text{ rad/s} \Rightarrow \underline{u}_2 = 4 \angle 30^\circ, |H(j4)| = 1, \angle H(j4) = 0^\circ$

$$\underline{y}_2 = \underline{u}_2 (1 \angle 0^\circ) = 4 \angle 30^\circ$$

• $\omega = 6 \text{ rad/s} \Rightarrow \underline{u}_3 = 7 \angle 45^\circ, |H(j6)| = 0, \angle H(j6) = 0^\circ$

$$\underline{y}_3 = \underline{u}_3 (0 \angle 0^\circ) = 0 \angle 45^\circ$$

Superimpose $\underline{y}_1, \underline{y}_2, \underline{y}_3$ to get $y(t)$:

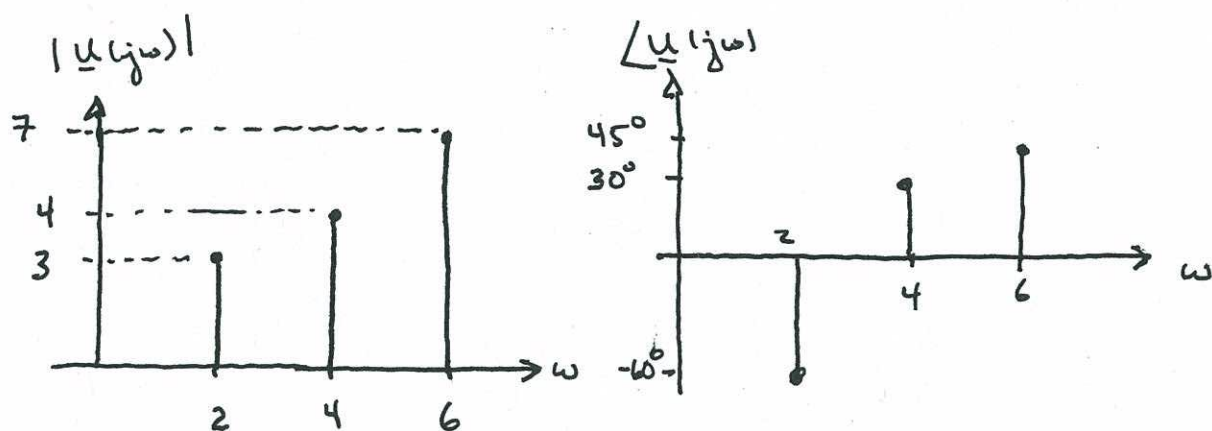
$$y(t) = 0 \cos(2t - 60^\circ) + 4 \cos(4t + 30^\circ) + 0 \cos(6t + 45^\circ)$$

$$\underline{\underline{y(t) = 4 \cos(4t + 30^\circ)}}$$

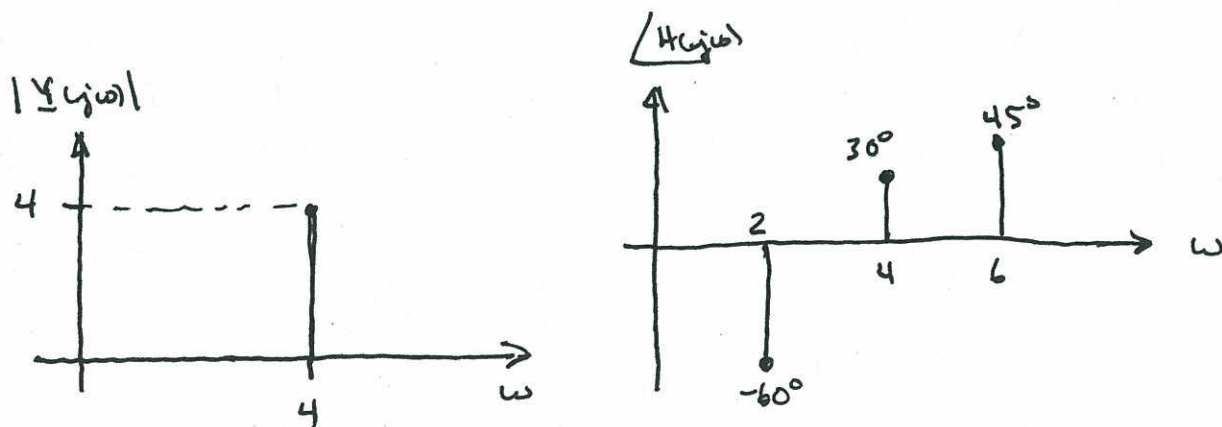
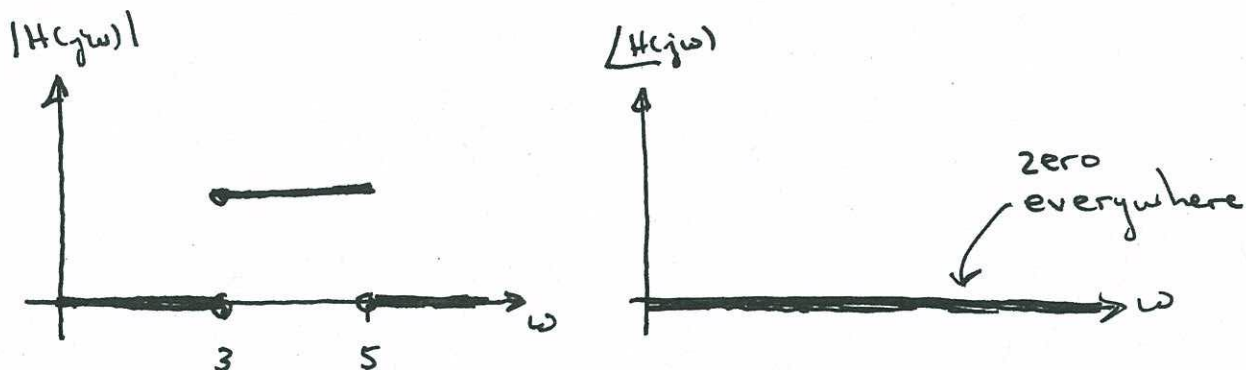
Exercises
Chapter 11.2

1.

(a)

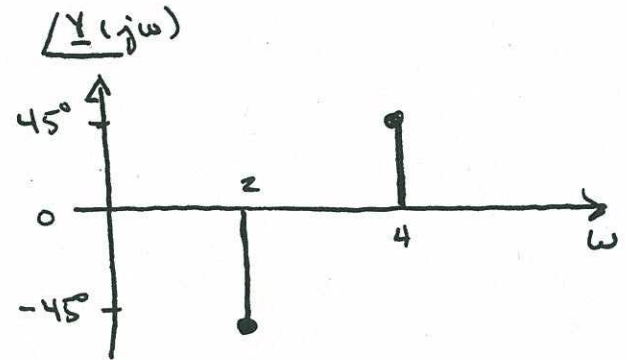
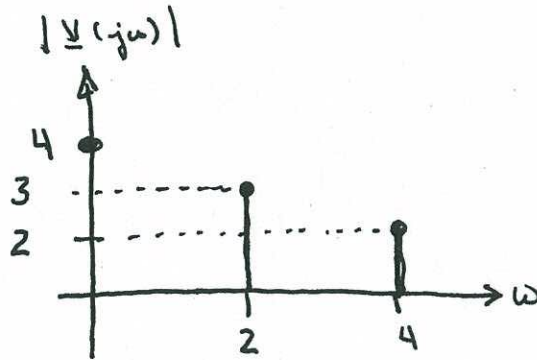


(b)



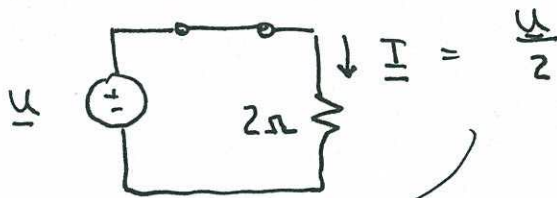
Exercises Chapter 11.2

2. $v(t) = 4 + 3 \cos(2t - 45^\circ) + 2 \cos(4t + 45^\circ)$



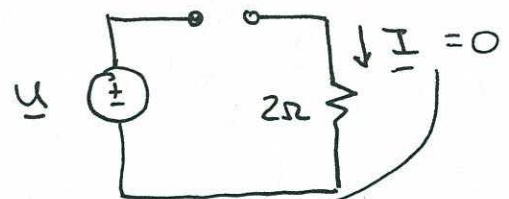
3.

$\omega \rightarrow 0$, inductor shorts:



$$|I| = \frac{1}{2}$$

$\omega \rightarrow \infty$, inductor opens



$$|I| = 0$$

Both check ✓

4.

$\omega \rightarrow 0$:



$$Y = \infty$$

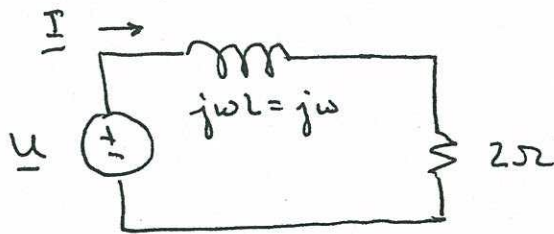
$\omega \rightarrow \infty$:



$$Y = 0$$

Exercises
Chapter 11.3

1. Frequency domain circuit:



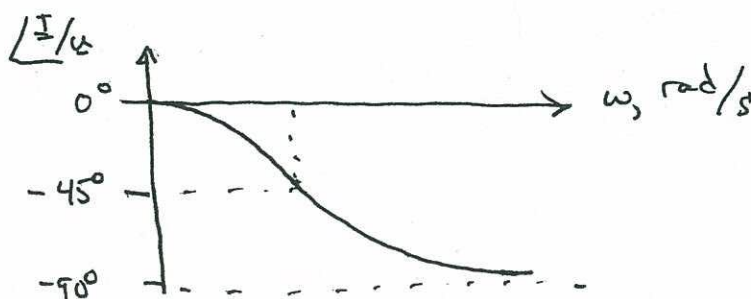
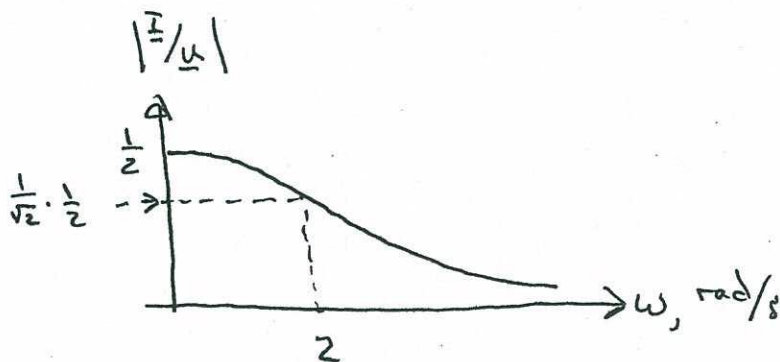
$$\underline{I} = \frac{\underline{U}}{2+j\omega} \Rightarrow \underline{\frac{I}{U}} = \frac{1}{2+j\omega} \Rightarrow \begin{cases} \left| \frac{I}{U} \right| = \frac{1}{\sqrt{2^2 + \omega^2}} \\ \angle \frac{I}{U} = -\tan^{-1}\left(\frac{\omega}{2}\right) \end{cases}$$

$$\left| \frac{I}{U} \right|_{\max} = \frac{1}{2} \quad (\text{when } \omega = 0)$$

$$\underline{\omega_c = 2 \text{ rad/sec}} \Rightarrow \left| \frac{I}{U} \right|_{\omega=\omega_c} = \frac{1}{\sqrt{2^2 + 2^2}} = \frac{1}{\sqrt{8}} = \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

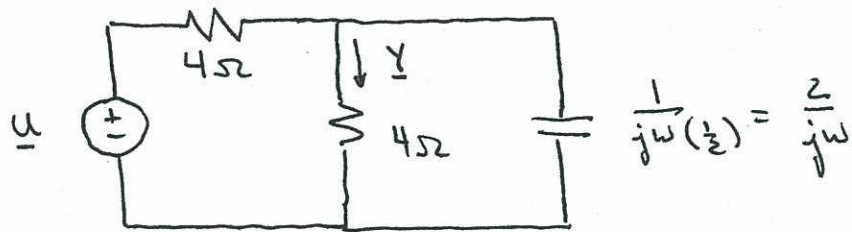
$\frac{1}{\sqrt{2}}$ times maximum value.

2.

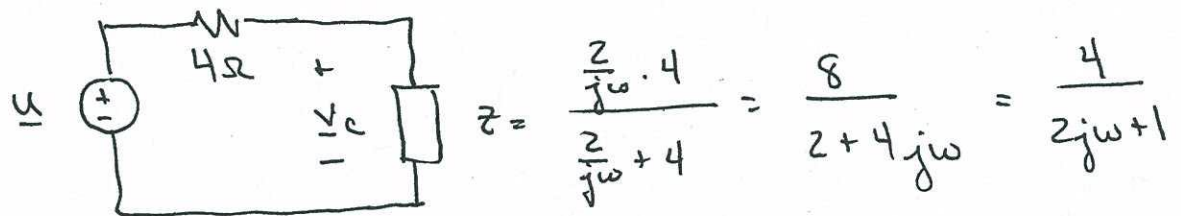


Exercises
Chapter 11.3

3. Frequency domain circuit:



Equivalent circuit:



V_c from voltage divider: $V_c = U \left[\frac{\frac{4}{2j\omega + 1}}{4 + \frac{4}{2j\omega + 1}} \right]$

$$V_c = U \frac{4}{8 + j8\omega} = \frac{1}{2 + j2\omega} U$$

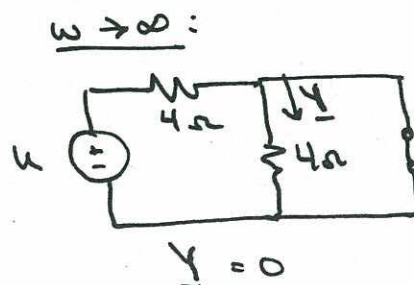
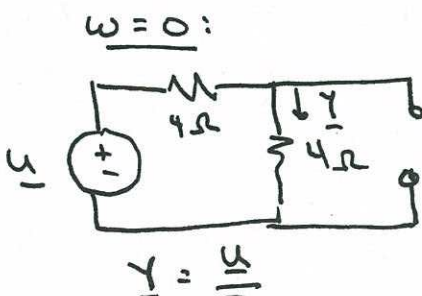
$$Y = \frac{V_c}{4\Omega} \Rightarrow Y = \frac{1}{8 + j8\omega} U \Rightarrow \frac{Y}{U} = \frac{1}{8 + j8\omega}$$

$\omega_c = 1 \text{ rad/s}$

Time constant:

$R_{eq} = 2\Omega$, $\tau = R_{eq} C = (2\Omega)(\frac{1}{2}F) = \underline{1 \text{ sec}}$

$\omega_c = \frac{1}{\tau}$



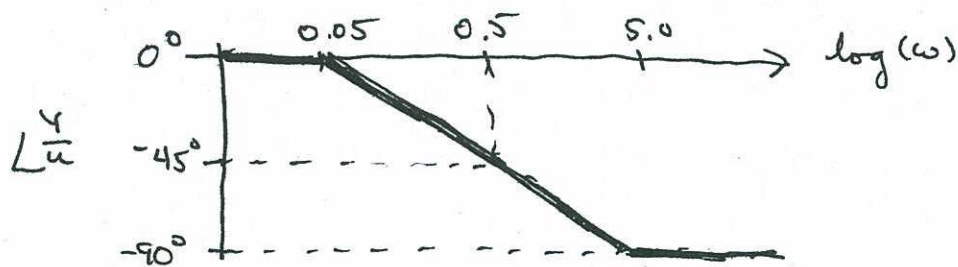
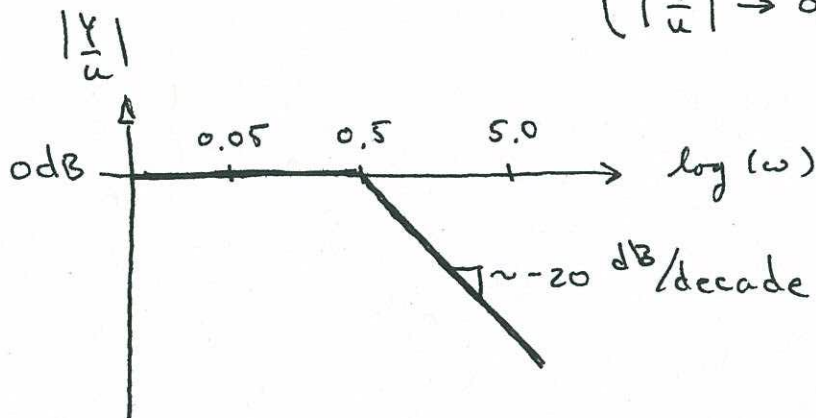
Low pass
Filter.

Exercises Chapter 11.4

1.

$$\frac{Y}{u} = \frac{1}{1+j2\omega}$$

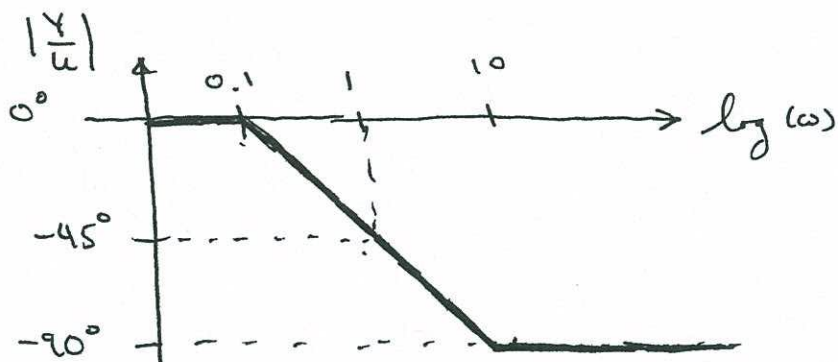
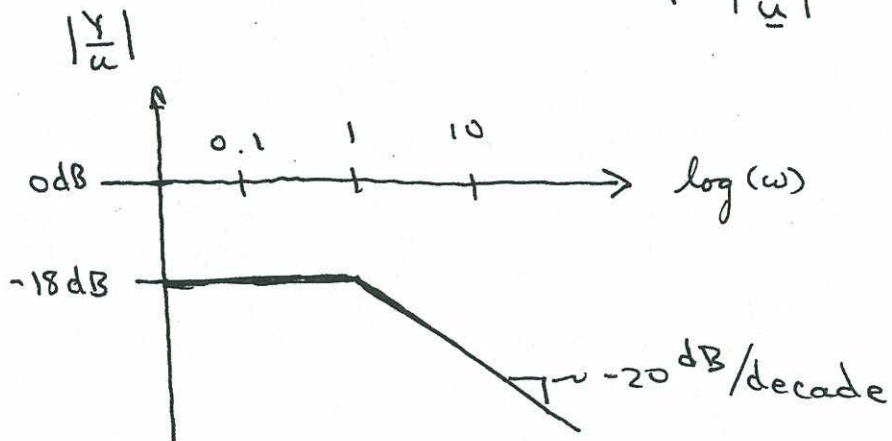
$$\Rightarrow \begin{cases} \text{DC gain} = 1 = 0 \text{ dB} \\ \omega_c = \frac{1}{2} \text{ rad/s} \\ \left| \frac{Y}{u} \right| \rightarrow 0 \text{ as } \omega \rightarrow \infty \end{cases}$$



Exercises
Chapter 11.4

2.

$$\frac{Y}{U} = \frac{1}{8 + j8\omega} \Rightarrow \begin{cases} \text{DC gain} = \frac{1}{8} = -18 \text{ dB} \\ \omega_c = 1 \text{ rad/sec} \\ \left| \frac{Y}{U} \right| \rightarrow 0 \text{ as } \omega \rightarrow \infty \end{cases}$$



Exercises
chapter 12.1

1. $i(t) = 2 \cos(100t) \Rightarrow \underline{I} = 2 \angle 0^\circ$

$$v(t) = 120 \cos(100t + 65^\circ) \Rightarrow \underline{V} = 120 \angle 65^\circ$$

$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i)$$

$$= \frac{(120)(2)}{2} \cos(65^\circ - 0^\circ)$$

$$= \underline{\underline{50.7 \text{ W}}}$$

Exercises
Chapter 12.2

1. $i(t) = 2 \cos(100t)$

$$v(t) = 120 \cos(100t + 65^\circ)$$

$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) = \frac{(120)(2)}{2} \cos(65^\circ - 0^\circ)$$

$$\underline{\underline{P = 50.7 \text{ W}}}$$

$$Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) = \frac{(120)(2)}{2} \sin(65^\circ - 0^\circ)$$

$$\underline{\underline{Q = 108.76 \text{ VAR}}}$$

Exercises

Chapter 12.3

1. $i(t) = 2 \cos(100t)$

$$\underline{\underline{I_{RMS} = \frac{2}{\sqrt{2}}}}$$

$$v(t) = 120 \cos(100t + 65^\circ)$$

$$\underline{\underline{V_{RMS} = \frac{120}{\sqrt{2}}}}$$

Exercises
chapter 12.4

1. $v(t) = 120 \cos(100t + 65^\circ)$

$$i(t) = 2 \cos(100t)$$

$$\text{Apparent power} = \frac{(120)(2)}{2} = \underline{\underline{120 \text{ VA}}}$$

$$\text{pf} = \cos(\theta_v - \theta_i) = \cos(65^\circ - 0^\circ) = \underline{\underline{0.42}}$$

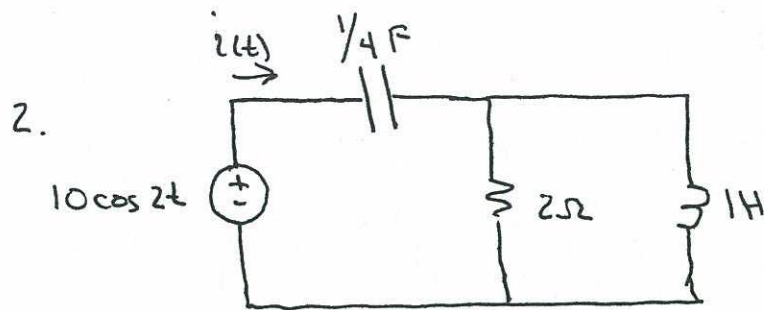
2. $P = V_{\text{RMS}} I_{\text{RMS}} \cos(\theta_v - \theta_i)$

$$100,000 \text{ W} = V_{\text{RMS}} (256 \text{ A}) (0.85)$$

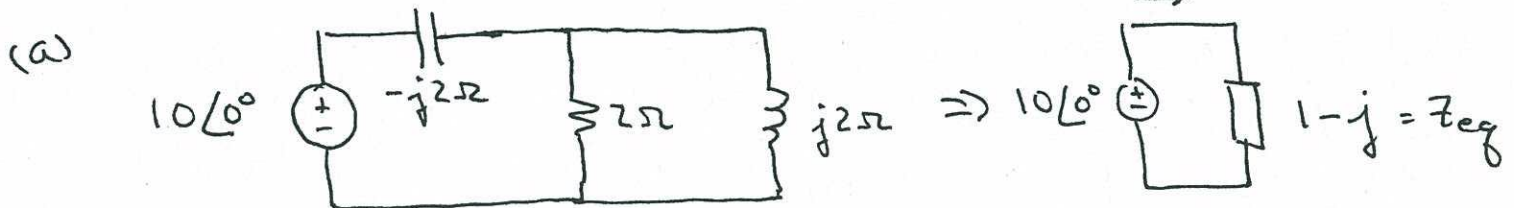
$$V_{\text{RMS}} = \frac{100,000 \text{ W}}{(256 \text{ A}) (0.85)} = \underline{\underline{460 \text{ V}}}$$

3. $\theta = \cos^{-1}(0.9) = \underline{\underline{25.8^\circ}}$

Exercises
Chapter 12.5



↓ Impedance domain circuit



$$\underline{I} = \frac{10 \angle 0^\circ}{1 - j} = \frac{10 \angle 0^\circ}{\sqrt{2} \angle -45^\circ} = \frac{10}{\sqrt{2}} \angle 45^\circ$$

$$\underline{i(t)} = \underline{\frac{10}{\sqrt{2}} \cos(2t + 45^\circ)}$$

(b)

$$S = \frac{V \underline{I}^*}{2} = \frac{(10 \angle 0^\circ) \left(\frac{10}{\sqrt{2}} \angle -45^\circ \right)}{2}$$

$$\underline{S = \frac{50}{\sqrt{2}} \angle -45^\circ \text{ VA}}$$

Note: complex conjugate change sign on phase angle

(c)

$$P = \frac{V I}{2} \cos(\theta_v - \theta_i)$$

$$= \frac{(10) \left(\frac{10}{\sqrt{2}} \right)}{2} \cos(-45^\circ)$$

$$\underline{P = 25 \text{ W}}$$